

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

B.Sc. in Software Engineering

Software Requirements Specification

for

IUT Medical Center

Agile / User Story Edition

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Version Information

Version	Date	Description	Author
1.0	2026-08-02	Initial release: Patient Care and Medical Record Management	Patient Care & Medical Record Management
1.1	2026-08-04	Integrated Pharmacy, Inventory and Laboratory scope	Pharmacy, Inventory & Laboratory Management
1.2	2026-08-05	Integrated Financial, Billing and Reimbursement scope	Financial, Billing & Reimbursement Management
1.3	2026-08-06	Integrated Emergency, Diagnostic and Referral scope and system-level structure	Emergency, Diagnostic & Referral Services
1.4	2026-08-08	Integrated the detailed Administrative, Reporting and Scheduling scope	Administrative, Reporting & Scheduling System
1.5	2026-08-09	Refined cross-system handoffs, diagnostic/laboratory routing, and NFR presentation.	Integrated SRS Team
1.6	2026-08-15	Aligned roles, emergency/ambulance flow, six-bed observation, prescription fields, reimbursement form, laboratory documents, CMO routing, and production UI behavior with the finalized wireframe.	Integrated SRS Team

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1. Introduction

1.1 Purpose

The purpose of this project is to develop one organized information system for the IUT Medical Center. The proposed system replaces scattered paper records, separate files, and manual communication with secure, role-based, searchable, and traceable workflows. Authorized users will be able to record, update, retrieve, and review the information required for daily clinical, emergency, diagnostic, pharmacy, laboratory, financial, administrative, reporting, and scheduling activities. The system supports Medical Center staff and other authorized IUT users but does not replace medical judgment, financial authorization, or official institutional approval.

This SRS consolidates the uploaded system-level draft and the supplied requirements for all five subdomains into one structured agile specification. It defines the requirements that guide design, development, testing, integration, and acceptance of the IUT Medical Center Support System, including detailed administrative duty scheduling, night on-call operations, diagnostic service reporting, and yearly policy-adherence auditing.

1.2 Document Conventions

- All identifiers use one global sequence across the complete system. Epics are written as EPIC-n, user stories as US-n.m, nonfunctional requirements as NFR-nn, business rules as BR-nn, and open items as TBD-nn. Numbering never restarts for a new subdomain.
- The word “shall” identifies a mandatory requirement; “should” identifies an important requirement that may be deferred; and “may” identifies an optional or permitted behavior.
- User stories follow the form: “As a [role], I want [goal], so that [benefit].”
- Every detailed story uses the same boxed table format with ID, Epic, Story, Priority, Precondition, Postcondition, Story Points, Dependencies, and Definition of Done. The Story Points field is retained but left blank in Version 1.6 pending estimation.
- Acceptance criteria appear immediately after the user story they verify. Each acceptance table uses three columns: Story ID, AC ID, and Acceptance Criteria. Within the Acceptance Criteria cell, Given, When, and Then are written on separate lines. Conservative criteria derived only from supplied story information remain marked “[Derived]”.
- NFRs follows a section-specific table structure. Performance requirements use ID, Requirement, and Metric / Target; safety requirements use ID, Safety Requirement, and Safeguard; and security requirements use ID and Requirement. A Mapped User Stories column is added to each NFR table for traceability.
- Dates use YYYY-MM-DD and time uses the 24-hour format. Shared capabilities are stated once and referenced as dependencies rather than duplicated.
- Any role-selection control shown in the interactive wireframe is for presentation/demo purposes only. In the production system, the authenticated user’s authorized role is obtained from institutional identity/authorization data and the correct dashboard loads automatically; the user does not choose a role from a combo box.

- Items listed in a backlog but missing detailed source text are not invented; they are recorded in Appendix B as source gaps.

1.3 Intended Audience and Reading Suggestions

Reader Type	Suggested Reading
Course Instructor / Evaluator	Sections 1–6 and the appendices.
Medical Center and ICT Stakeholders	Sections 1–3, the relevant subdomain subsection in Section 4, and Section 5.
Developers	All sections, especially detailed stories, interfaces, dependencies, and nonfunctional requirements.
Testers	Story acceptance criteria, Section 5, and the source-gap register.
Clinical Staff	Product scope, user classes, and the Patient Care & Medical Record Management clinical workflow stories in Section 4.2.
Finance / Administration Staff	Financial, Billing & Reimbursement Management and Administrative, Reporting & Scheduling System scope, including duty rosters, shift changes, absence/cover, night on-call operations, diagnostic activity reporting, policy-adherence audit, interfaces, business rules, and nonfunctional requirements.

1.4 Product Scope

The product is an integrated web-based information system for the IUT Medical Center. It will maintain longitudinal patient records; support consultations, nursing care, prescriptions, emergency triage, referrals, diagnostic requests and results; manage medicine dispensing, inventory, expiry, procurement, billing, reimbursement, and payments; and provide administration, scheduling, notification, reporting, monitoring, and audit capabilities. Printable referral letters, certificates, receipts, and reports required by IUT procedures will remain supported.

Area	Subdomain	Primary Responsibility
Clinical Care	Patient Care & Medical Record Management	Identity-linked patient records, clinical encounters, nursing operations, prescriptions, laboratory requests, certificates, and emergency-examination services.

Area	Subdomain	Primary Responsibility
Emergency Services	Emergency, Diagnostic & Referral Services	Emergency registration and triage, direct doctor/nurse ambulance requests, driver acceptance and live map tracking, six-bed Bed Rest / Observation, referrals, diagnostic booking/execution, payment coordination, and diagnostic reporting.
Pharmacy & Laboratory	Pharmacy, Inventory & Laboratory Management	Medicine availability and dispensing, stock/expiry control, inventory reports and procurement requests, laboratory request processing and result publication.
Finance & Procurement	Financial, Billing & Reimbursement Management	Medical-expense reimbursement, referral billing, accounts/payment execution, audit trails, and procurement lifecycle management.
Administration	Administrative, Reporting & Scheduling System	Weekly and night on-call duty rosters, shift-change decisions, absence/cover restoration, staff/public schedule visibility, night-call logging and escalation, formal laboratory service-count reporting, notifications, and related administrative reporting.

1.5 Sources and Existing Resources

The specification is based on the supplied subsystem requirements, the existing institutional formats, and the records currently maintained by the Medical Center. Resource descriptions are intentionally concise; detailed requirements are defined in the relevant functional sections.

Ref.	Source / Resource	Use in This SRS
SRC-1	Integrated IUT Medical Center SRS draft	System scope, conventions, interfaces, shared requirements and document structure.
SRC-2	Patient Care & Medical Record Management requirements	Authentication/session, patient care, prescription, laboratory-request and emergency-examination workflows.
SRC-3	Emergency, Diagnostic & Referral requirements	Emergency, referral, diagnostic workflows, criteria, NFRs and rules.
SRC-4	Pharmacy, Inventory & Laboratory requirements	Dispensing, inventory, procurement reporting and laboratory processing.
SRC-5	Financial, Billing & Reimbursement requirements	Claims, referral billing, payment, audit and procurement.
SRC-6	SRS format reference	Heading hierarchy, boxed stories, neutral styling and compact presentation.

Ref.	Source / Resource	Use in This SRS
SRC-7	Administrative, Reporting & Scheduling requirements	Duty rosters, shift changes, schedule visibility, night on-call, reporting and yearly policy audit.
SRC-8	Pre-existing prescription format(s)	Presentation reference for prescriptions produced by the proposed system.
SRC-9	Current-workflow record inventory	Authoritative list of the records maintained in the current workflow and used to reconcile the relevant requirements.

1.5.1 Current-Workflow Records Used as Requirement Inputs

The provided current-workflow image identifies all records presently tracked by the Medical Center. These records are used as evidence for the corresponding requirement areas; the proposed system may structure them digitally without losing their required information.

Current Record	Relevant Requirement Area / Stories
Daily Medicine Stock	Inventory stock, dispensing and expiry management: EPIC-14 and EPIC-15.
Monthly Medicine Bill	Monthly medicine-bill reporting and financial review: US-2.5 and related finance workflows.
Emergency Procedure Record	Patient-care activity and emergency workflows: US-2.1 and EPIC-6–EPIC-8.
Dressing Procedure Record	Nursing / medical-attendant patient-care activity: US-2.1.
Referral Record	Referral initiation, documentation, payment and follow-up: EPIC-9 and EPIC-10.
Prescription Record	Prescription creation, finalization, active viewing and history: EPIC-3; dispensing consumes the final prescription.
Lab Reports	Formal laboratory request, result verification, publication, document preview, and authorized download workflow: EPIC-17; only verified/published reports are available as final downloadable documents and are linked back to the unified patient record.

2. Overall Description

2.1 Product Perspective

The IUT Medical Center Support System is a new integrated product that digitizes current Medical Center workflows and connects to the IUT ICT Center identity/account services and approved communication services. The ICT Center remains responsible for institutional identity and account provisioning, including official IUT identity, email, category, account status and dependent relationships. Creating or registering institutional user accounts is outside the Medical Center system boundary. The Medical Center system begins with authentication of ICT-provisioned users and owns the operational records produced by its clinical, emergency, diagnostic, pharmacy, laboratory, financial, administrative, reporting and scheduling workflows.

The five subsystems interoperate through controlled interfaces. Patient Care owns the unified clinical record; Emergency, Diagnostic & Referral Services owns emergency/referral and its diagnostic-service flow; Pharmacy, Inventory & Laboratory Management owns dispensing, inventory, and formal laboratory processing; Financial, Billing & Reimbursement Management owns monetary execution, financial audit/policy-adherence review, and procurement after CMO submission; and Administrative, Reporting & Scheduling System owns duty scheduling, night-call logging, and aggregate administrative reporting. Nurses/medical attendants may submit permitted operational records or reports to the CMO; the CMO routes the submitted item to the appropriate department while preserving the original record and routing history. Cross-subsystem actions reuse the owning workflow rather than create duplicate records or logic. Final clinical, laboratory, dispensing, referral, and financial records are corrected through controlled amendments or adjustments rather than destructive editing.

2.2 Product Functions

- Authenticate ICT-provisioned users, enforce role-based authorization, support first-time password change where flagged, and manage secure sessions and logout.
- Retrieve eligible student, faculty, staff, and dependent information and maintain longitudinal medical records.
- Record consultations, clinical assessments, vital signs, nursing procedures, prescriptions, clinical amendments, medical certificates, and emergency-examination requests. When a doctor starts a consultation, the workspace shall first show the patient profile, alerts/allergies, latest vitals, previous consultations, active/past prescriptions, referrals, emergency history, and published laboratory/diagnostic results available to that doctor.
- Register and triage emergencies, manage a separate six-bed Bed Rest / Observation board, allow the on-duty doctor or nurse to issue an ambulance request directly to the on-duty driver, record driver acceptance/rejection, display the accepted driver's live map-based location/status to the requesting clinical staff, and manage referrals to approved hospitals or specialists.
- Manage the diagnostic-service workflow owned by Emergency, Diagnostic & Referral Services: check service availability, collect permitted upfront payments, perform supported on-site diagnostic services, record diagnostic results, and recommend approved external providers when the service cannot be completed on site.
- Manage the separate laboratory workflow owned by Pharmacy, Inventory & Laboratory Management: receive doctor-issued laboratory requests, process laboratory tests, verify findings, and publish final laboratory reports

to the unified patient record.

- Display medicine availability, retrieve prescriptions, dispense medicines, update inventory atomically, manage batches, stock, expiry, alerts, and inventory reports.
- Generate charges and bills, validate reimbursement and referral claims, record approvals, execute payments and adjustments, and maintain immutable financial audit records. Accounts shall use the consolidated payment/status workspace rather than a separate duplicate finance screen.
- Manage procurement requests, approval, tenders, vendor selection, delivery recording, and inventory updates.
- Plan, validate, publish, and display weekly duty rosters and night on-call rosters; prevent overlapping or uncovered published shifts; and notify staff whose own duty changes.
- Manage shift-change requests, CMO decisions, absence and lateness records, same-role replacement, and escalation of unfilled shifts.
- Provide read-only personal schedules and a public on-duty board, record night hotline calls, and hand night cases requiring transfer to the existing Emergency, Diagnostic & Referral ambulance workflow rather than creating a second ambulance process.
- Generate and retain aggregate blood/urine laboratory service-count reports from EPIC-17 without duplicating individual laboratory results, and allow Accounts / Finance users with audit permission to run the yearly reimbursement-policy adherence audit from authorized financial records.
- Provide the shared configuration, notification, export, monitoring, and audit services required by the documented workflows and nonfunctional requirements.

2.3 User Classes and Characteristics

The production application uses role-based access derived from the authenticated institutional account. The role selector shown in the interactive wireframe is only a presentation aid and is not part of the production interface. After login, the system automatically opens the dashboard/workspace authorized for the user.

Persona / Role	Characteristics and Main Functions
Student / Patient	Views permitted health records, prescriptions, published laboratory/diagnostic results, referrals and notifications; submits medical-certificate, emergency-examination and eligible reimbursement requests; and tracks their status. Student identity uses the authoritative IUT student ID.
Faculty / Staff	Receives care, views permitted own/dependent health information, submits eligible requests/claims, and uses a faculty/staff identity distinct from the student-ID format. Eligible dependents remain linked through the primary member.

Persona / Role	Characteristics and Main Functions
Doctor / Medical Officer	Searches patients, begins consultations with prior patient context already visible, records clinical assessment, creates/finalizes prescriptions, requests investigations, reviews published results, creates referrals/certificates, makes emergency clinical decisions, and may call an ambulance directly when on duty. After driver acceptance, the doctor can view the driver's live map-based location and trip status.
Nurse / Medical Attendant	Records routine vitals once, performs triage-specific emergency assessment without re-entering the same vitals, records emergency/dressing procedures, manages six-bed Bed Rest / Observation, daily tracking, pharmacy/inventory operations where authorized, and Records & Reports. May call an ambulance directly and view the accepted driver's live location/status. May review and submit permitted operational records/reports to the CMO for routing.
Laboratory Technician	Receives doctor-issued laboratory requests, processes tests, records findings, verifies results, publishes final laboratory report documents, and maintains published-report history. Only verified/published reports are available for authorized View Report / Download Document actions.
CMO	Provides clinical/administrative oversight, maintains duty/on-call coverage, decides shift-change/coverage matters, reviews reimbursement and external-referral claims, receives nurse/medical-attendant records/reports, selects the appropriate destination department and forwards them with traceable routing, reviews procurement medical need, and handles emergency escalation.
Registrar	Reviews emergency-examination requests and schedules approved examinations, records staff absence/lateness and coordinates administrative scheduling handoffs to the CMO.
Ambulance Driver	Receives ambulance call requests created directly by an authorized on-duty doctor or nurse, accepts or rejects the request, updates trip state and live location, and completes transfer history. The driver does not make the clinical transfer decision.
Accounts / Finance / Cashier	Uses one consolidated Finance workspace that separates approved payouts, referral counter payments, external-hospital bills/claims and refunds; executes payment instructions, updates payment status, views immutable audit trails, and performs authorized policy-adherence review.
Procurement / Authorized Admin / VC	Receives approved procurement information, performs the required approval/tender/vendor/delivery steps according to permission, and primarily updates controlled process status/decisions rather than duplicating upstream clinical or inventory forms.
Public On-Duty Board	Non-login, read-only access limited to current duty/on-call information, shift timing and the approved Medical Center hotline; no patient or clinical information is exposed.

Persona / Role	Characteristics and Main Functions
External Supporting Actors	The IUT ICT Center provides authoritative identity/account data; the Transport Office provides read-only driver-duty information; external hospitals/specialists/diagnostic labs exchange approved documents through authorized channels. These are integrations/supporting actors, not selectable Medical Center application roles.

User Class Overview Diagram

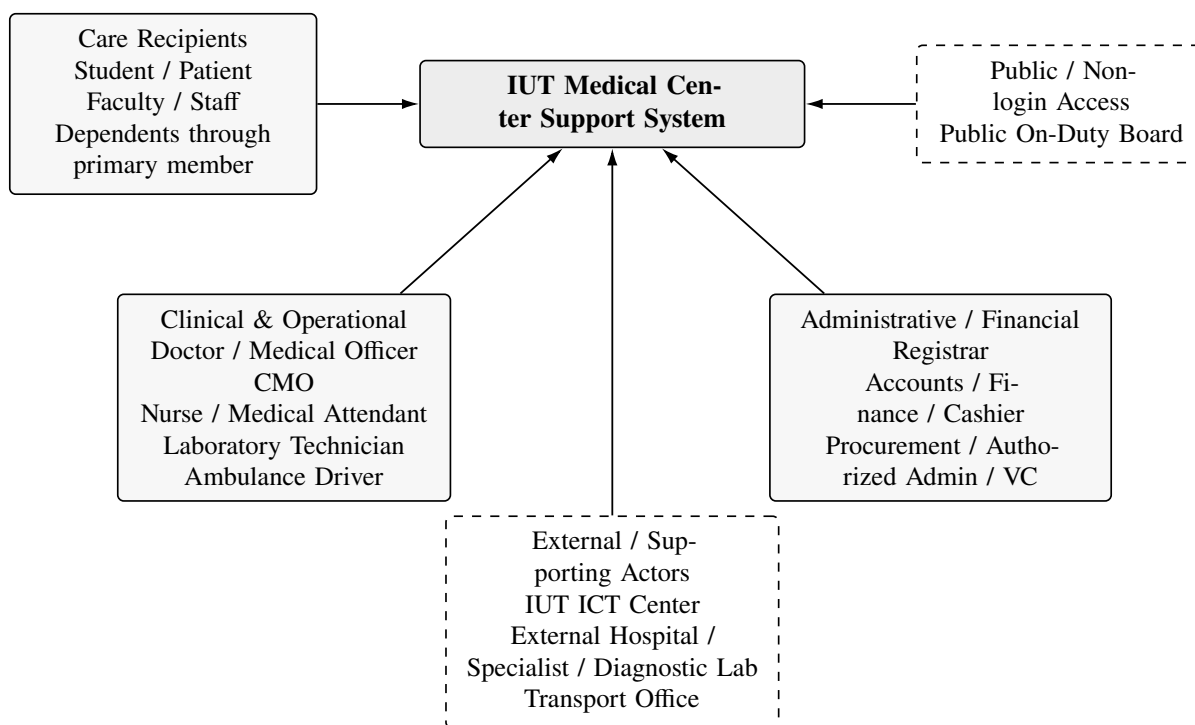


Figure 1: Production user classes and external supporting actors. The wireframe role switcher is demonstration-only; production routing is automatic after login.

2.4 Operating Environment

- Current desktop and mobile web browsers on IUT-managed or approved devices.
- Application, database, file storage, backup, and monitoring infrastructure approved by the IUT ICT Center.
- Secure network connectivity for normal operation and support for standard institutional printers and document scanners.
- Coexistence with the IUT ICT identity database, official email service, approved SMS/export services, centralized medical records, and approved banking interfaces.

- Emergency and operational workflows must remain available continuously; the public on-duty board and night hotline information shall remain readable when protected clinical functions are not in use; on-site diagnostic operations follow the stated 08:00–14:00 service window.

2.5 Design and Implementation Constraints

- The IUT ICT Center database is authoritative for institutional identity and eligibility; ICT-controlled fields shall not be silently overwritten.
- The Medical Center application shall not create institutional user accounts; account provisioning and authoritative identity/account status are external IUT ICT Center responsibilities consumed through an approved interface.
- Access shall be role-based and least-privilege. Sensitive medical, referral, laboratory, and financial data shall not appear in general reports or notification bodies.
- All financial transactions and material clinical/operational changes require immutable or append-only audit records.
- Required printable referral, certificate, receipt, and report formats shall be preserved even when the workflow is digital. Existing approved prescription format(s) shall be used as the presentation reference unless a replacement format is formally approved.
- Inter-subsystem integration shall use documented, secure interfaces. Payment processing shall defer to the approved banking/payment layer. The Emergency ambulance workflow may read the external Transport Office driver-duty schedule through an approved read-only interface; the on-duty doctor or nurse issues the ambulance request directly to the eligible driver. After driver acceptance, approved live location/status updates are visible to the requesting doctor/nurse. Administrative, Reporting & Scheduling hands night transfer cases to this same Emergency request flow and does not maintain a duplicate ambulance process.
- The public on-duty board shall expose only current duty/on-call information, shift timing, and the approved hotline information; it shall not expose patient or clinical details.
- Final technology stack, production hosting, retention periods, external report templates, and several API contracts remain subject to stakeholder and ICT approval.

2.6 User Documentation

- Role-based quick-start guides for patients, doctors, nurses, pharmacy, laboratory, finance, registrar, procurement, and administrators.
- Online field help, status explanations, validation messages, process guidance, and interactive tooltips within major forms.
- Emergency triage quick-reference guidance and referral/diagnostic counter procedures.
- Administrative, reporting, financial, and procurement operating guides.
- Technical runbook for configuration, integrations, monitoring, backup, recovery, and audit review.

2.7 Assumptions and Dependencies

2.7.1 Assumptions

1. The IUT ICT Center is the authority for institutional user records and account provisioning; users who require protected Medical Center access are provisioned by ICT before they attempt login.
2. Students, faculty, staff, doctors, nurses, medical attendants, laboratory staff, CMO personnel, finance staff, registrar, finance, procurement/authorized administration users, and ambulance drivers have valid institutional records before protected access is requested.
3. Faculty and staff dependent records normally exist and are linked to the corresponding primary member; approved exceptional entry rules apply where necessary.
4. Login-enabled users have valid official IUT email addresses and approved communication services are available.
5. Medicine, laboratory test, service, fee, staff, room, partner hospital, external lab, and policy master data are approved before production use.
6. External hospitals, specialists, and diagnostic labs do not receive direct Medical Center system access in the current scope; information and documents are exchanged through approved channels and authorized IUT users record required external submissions in the system.
7. Financial limits, reimbursement rules, retention periods, and examination policies are provided by responsible IUT authorities.
8. The Medical Center operates continuously, emergency care is not delayed by incomplete identity data, and emergency care is free of cost.
9. Referral and diagnostic services follow the supplied “Pay First, Get Refunded Later” rule unless an approved exception applies.
10. The exact on-site diagnostic test list, emergency hotline, live-location technology/provider, and some external integrations remain to be confirmed. The Bed Rest / Observation capacity is fixed at six beds for this system scope.
11. Staff lists, shift timings, on-call eligibility, staff contact details and preferred notification channels are configured before roster publication; a published day roster exists before the corresponding night on-call roster is finalized.
12. The approved reimbursement policy and authorized reimbursement records required for the yearly adherence audit are available through controlled read access only to an explicitly Accounts / Finance Officer with audit permission role.

2.7.2 Shared Dependencies

Dependency	Purpose / Boundary	Owner	Related User Stories
Institutional Identity and Account Provisioning	Authoritative identity, official email, category, account status, dependent relationships and initial account provisioning. The Medical Center system consumes these data but does not create institutional accounts.	IUT ICT Center	US-1.1–US-1.4; US-2.1–US-5.3; US-6.2; protected stories requiring institutional identity.
Authentication and Session Management	Login, first-time password change, role validation, session expiry and logout for ICT-provisioned users.	Medical Center / shared security	US-1.1–US-1.4; all protected stories.
Unified Patient and Medical Record	Valid patient profile, clinical history, allergies, prescriptions, referrals, laboratory links and certificates.	Patient Care & Medical Record Management	US-2.1–US-5.3; US-6.1–US-13.1; relevant pharmacy/lab stories.
Medicine Inventory and Laboratory Operations	Medicine availability, stock deduction, batch/expiry, formal laboratory request/result processing, laboratory verification, and inventory reporting.	Pharmacy, Inventory & Laboratory Management	US-3.2–US-4.1; US-14.1–US-17.3; US-26.1–US-26.2.
Diagnostic Service Operations	Availability, booking/execution status, diagnostic result recording, external-provider recommendation, and clinical diagnostic activity reporting for the Emergency, Diagnostic & Referral service flow.	Emergency, Diagnostic & Referral Services	US-11.1–US-13.1.
Billing, Payment, Reimbursement and Procurement	Charges, claims, receipts, refunds, bank transfer, financial audit/policy-adherence review, approval, tender and delivery lifecycle.	Financial, Billing & Reimbursement Management	US-10.1–US-11.2; US-16.2; US-18.1–US-21.4; US-27.1–US-27.2.
Exceptional Refund Approval Authority	Provides the additional university-level authorization required for disputed or over-limit refunds. The exact responsible authority remains to be confirmed.	Designated higher financial authority (TBD)	US-10.3.
Staff Scheduling, Notification and Administrative Reporting	Duty/on-call rosters, shift changes, absence/cover, schedule views, notifications and formal laboratory service-count reports.	Administrative, Reporting & Scheduling System	US-22.1–US-26.2; notification-consuming stories across other subdomains.

Dependency	Purpose / Boundary	Owner	Related User Stories
Transport Office Driver-Duty Schedule and Driver Location Feed	The Transport Office duty schedule is read-only and identifies the eligible on-duty driver. After that driver accepts an ambulance request, the driver's approved mobile/GPS location and trip-status updates are exposed to the requesting doctor/nurse for map-based tracking. Night on-call escalation reaches the same request flow through US-25.2.	External Transport Office / Ambulance Driver	US-7.2–US-7.3; US-25.2 through the Emergency transfer handoff.
Secure Document Storage	Referral proof, certificates, bills, receipts, reports and supporting documents with controlled access/version preservation.	Shared service	US-2.3; US-5.1–US-5.3; US-9.2–US-10.3; US-17.2–US-17.3; reporting stories that store exports.
External Referral Claim Intake	Referral bills, claims, and supporting documents are received through an approved external channel and entered by authorized internal IUT staff; no external-hospital system login is provided.	Authorized IUT Finance / Medical Center staff	US-19.1–US-19.2.
External Services	Partner hospitals, external diagnostic labs, approved bank API, official email/SMS and printer/scanner interfaces.	External / ICT-approved	US-9.3–US-12.4; US-18.1–US-21.4; US-22.3; US-25.2.

3. External Interface Requirements

External Interface Overview Diagram

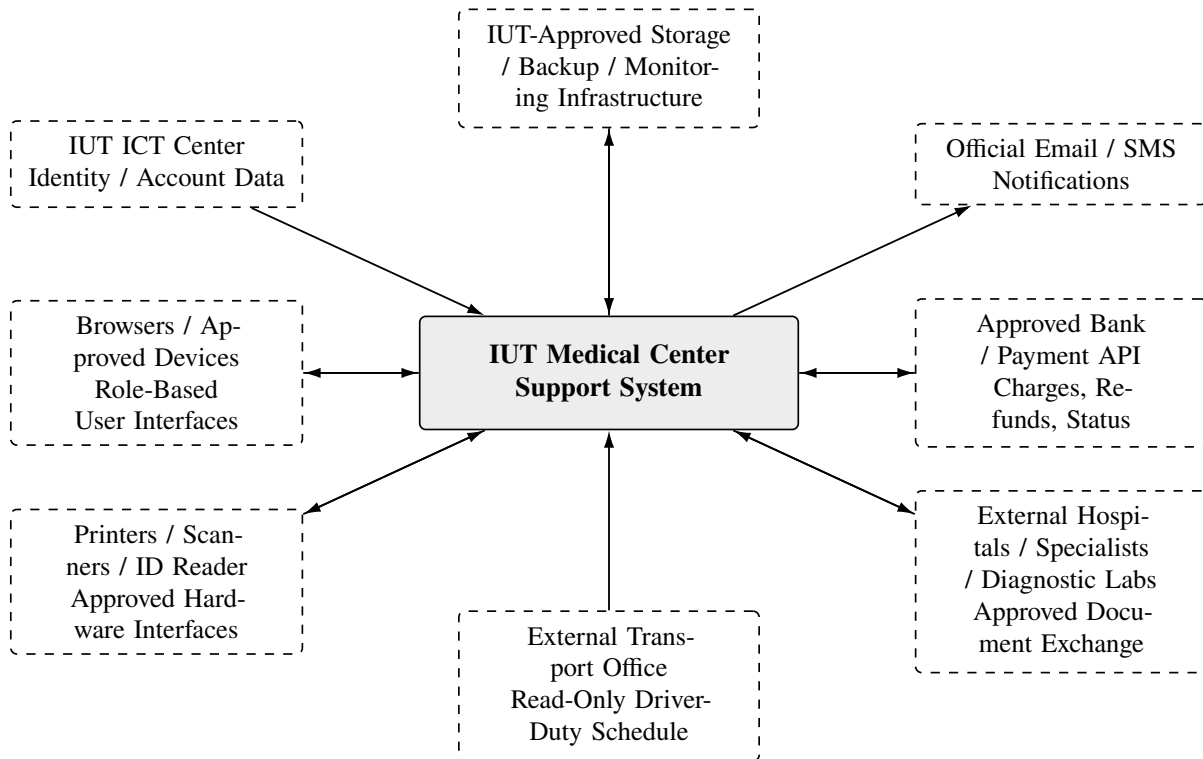


Figure 2: External interfaces surrounding the Medical Center system boundary.

3.1 User Interfaces

The system shall provide consistent role-based dashboards and responsive forms. In production, the authenticated role is resolved automatically and the correct dashboard opens without a role-selection combo box; any role selector in the wireframe is demonstration-only. Required fields shall be clearly marked; validation shall appear near the affected field; statuses shall use one governed vocabulary; normal progress changes shall use a status dropdown plus Update Status, while true decisions shall use explicit Accept/Reject actions. Record rows shall use a consistent View Details action that opens a structured read-only/detail view; generic duplicate Edit controls and redundant dashboard panels shall be avoided. Wide data tables shall scroll within their panel on narrow screens instead of compressing labels/actions into unreadable cells. Clinical diagnostic findings and laboratory/diagnostic results shall appear in one consolidated results area rather than a separate diagnosis panel. Emergency screens shall use high contrast, large touch targets, and only the minimum fields required to open and triage a case.

Interface	Primary Users	Main Elements
Patient Portal	Patients and students	Permitted history, prescriptions, diagnostics, referrals, certificates, exam requests, bills, claims, and notifications.
Clinical Workspace	Doctors, nurses, medical attendants, CMO	Patient search; pre-consultation patient context (alerts/allergies, previous visits, prescriptions, referrals, emergency history, latest vitals); encounter; treatment; prescription; consolidated Lab & Diagnostic Results; certificates; emergency care; separate six-bed Bed Rest / Observation; ambulance request and live driver map where authorized.
Pharmacy and Inventory	Nurse / Medical Attendant, CMO	Prescription retrieval, dispensing, medicine availability, stock ledger, batches, expiry, counts, alerts, records/reports, and procurement-need submission.
Laboratory Workspace	Lab staff, doctors, patients, nurses/medical attendants where authorized	Request queue, specimen/equipment status, processing, result entry, verification, publication, exceptions, history, report-document preview, and download of verified/published laboratory reports.
Finance and Procurement	Accounts / Finance / Cashier, CMO, Procurement / Authorized Admin / VC	One consolidated Finance workspace separating approved payouts, referral counter payment (money in), external-hospital bills/claims, and refunds (money back), plus payment status and immutable audit trail. Procurement views emphasize received request/tender/delivery information and controlled status/decision updates rather than duplicate editing forms.
Administration and Scheduling	CMO, Registrar, staff, Accounts / Finance / Cashier with audit permission	Weekly and on-call roster editor, clash/gap warnings, shift-change approval queue, absence/cover handoff, personal duty schedule, public on-duty board, night-call log and escalation, laboratory service-count reporting, yearly policy-adherence audit within Finance permission, notifications, dashboards and logs.

3.2 Hardware Interfaces

Hardware	Interaction
Standard computers, laptops, tablets, and smartphones	Web-based data entry, review, approval, monitoring, and patient access.
Printers	Referral letters, certificates, receipts, result slips, and reports.

Hardware	Interaction
Document scanners / cameras	Bills, referral proof, certificates, and supporting-document upload.
ID card reader / scanner	Reads institutional ID and passes it to the approved identity lookup interface.
Optional barcode / label devices	Future laboratory specimens or medicine-batch identification through documented interfaces.
Ambulance driver mobile/GPS location source	Provides live location/status for map-based monitoring after a driver accepts an ambulance request. The exact device/vendor/protocol remains TBD, but live tracking is in scope.

3.3 Software Interfaces

Interface	Data Exchanged	Requirement
IUT ICT Center Identity / Account Interface	Institutional ID, identity, official email, category, account status, authorization attributes, dependent relationship	Read/synchronized access through an approved interface; ICT provisions the institutional account and ICT-owned fields remain authoritative. The Medical Center application does not create institutional accounts.
Unified Patient Record API	Patient profile, history, allergies, encounters, prescriptions, referrals, tests, certificates	Role-controlled REST/JSON or ICT-approved equivalent with referential integrity.
Inventory / Laboratory API	Medicine availability, stock deductions, thresholds, batches, laboratory requests, verified laboratory results, specimen/equipment status	Supports the Pharmacy, Inventory & Laboratory Management workflow. Laboratory results are published only after the verification gate defined in EPIC-17.
Diagnostic Service Interface	Diagnostic-service availability, booking status, on-site execution status, diagnostic result references, and external-provider recommendation data	Supports EPIC-11–EPIC-13 as a service flow distinct from the formal laboratory workflow in EPIC-17.
Payment / Banking API	Charges, receipts, refunds, transfer instructions, acknowledgements, payment status	Secure approved protocol; no sensitive bank data exposed to unauthorized modules.

Interface	Data Exchanged	Requirement
Scheduling / Reporting / Audit API	Published duty and on-call rosters, shift-change/absence events, notification delivery results, EPIC-17 laboratory service-count records, authorized reimbursement records, generated report metadata, and night-transfer handoff references	Documented contracts, role-controlled reads, timestamped events, and immutable decision/audit metadata; night transfer is handed to the Emergency ambulance request workflow and EPIC-12 activity and individual laboratory results are not duplicated into the EPIC-26 administrative count report.
External Partner Communication	Referral letters, external hospital bills/claims, supporting documents, external diagnostic information, and acknowledgements	Approved external channels only. External hospitals and other partners do not receive general Medical Center system accounts; authorized internal users record required submissions.
Official Email / SMS	Status, result, claim, referral, vendor and administrative notifications	Approved templates; detailed sensitive records remain in the authenticated portal.
Secure Document Storage	Bills, referral proof, certificates, receipts, published laboratory report documents, reports, supporting documents	Type/size/readability validation, role access, version preservation, and malware controls.
Ambulance Driver / Live Location Interface	Ambulance request, Case ID, requester, driver acceptance/rejection, vehicle, current location, last update time, ETA when available, and trip status	The request is issued by an authorized on-duty doctor/nurse and delivered to the eligible driver. Live map data is exposed only after driver acceptance and only to authorized clinical users; exact mobile/GPS provider/protocol remains TBD.

3.4 Communications Interfaces

- Browser and service communication shall use HTTPS/TLS or another ICT-approved secure protocol.
- Application interfaces shall use documented request/response formats and consistent error codes without exposing internal secrets.
- Email and SMS shall use approved templates, a reference/status summary, and a link or instruction to use the authenticated portal for sensitive details. Duty-change notifications shall record delivery outcome; failed delivery shall be retried or sent through an approved alternate channel where configured.
- Background synchronization and notification jobs shall record the last successful run, failures, and retry count.
- File uploads shall validate approved file type, size, readability, and storage outcome.

4. Epics and User Stories

This section consolidates the detailed epics and stories supplied for all five subdomains. Each epic presents its description and priority first, followed by user stories; the acceptance criteria for each story appear immediately after that story. Epic, story, and acceptance-criterion identifiers follow one global sequence.

4.1 Integrated Story Map / Backlog Overview

Epic ID	Subdomain	Epic Name	Priority	Release	Story IDs
EPIC-1	Patient Care & Medical Record Management	Authentication and Session Management	High	Release 1	US-1.1, US-1.2, US-1.3, US-1.4
EPIC-2	Patient Care & Medical Record Management	Nurse and Medical Attendant Operations	High	Release 1 and Release 2	US-2.1, US-2.2, US-2.3, US-2.4, US-2.5
EPIC-3	Patient Care & Medical Record Management	Prescription Management	High	Release 1	US-3.1, US-3.2, US-3.3, US-3.4
EPIC-4	Patient Care & Medical Record Management	Laboratory Test Management	High	Release 2	US-4.1
EPIC-5	Patient Care & Medical Record Management	Emergency Examination Services	High	Release 2	US-5.1, US-5.2, US-5.3
EPIC-6	Emergency, Diagnostic & Referral Services	Emergency Registration & Identity Verification	High	Release 1	US-6.1, US-6.2, US-6.3
EPIC-7	Emergency, Diagnostic & Referral Services	Vital Signs, Triage & Ambulance Request / Live Tracking	High	Release 1	US-7.1, US-7.2, US-7.3
EPIC-8	Emergency, Diagnostic & Referral Services	Bed Rest / Observation Management	Medium	Release 1	US-8.1
EPIC-9	Emergency, Diagnostic & Referral Services	Referral Initiation & Documentation	High	Release 1	US-9.1, US-9.2, US-9.3, US-9.4
EPIC-10	Emergency, Diagnostic & Referral Services	Referral Payment & Refund Processing	High	Release 1	US-10.1, US-10.2, US-10.3

Epic ID	Subdomain	Epic Name	Priority	Release	Story IDs
EPIC-11	Emergency, Diagnostic & Referral Services	Diagnostic Test Booking & Payment	High	Release 2	US-11.1, US-11.2
EPIC-12	Emergency, Diagnostic & Referral Services	Diagnostic Service Execution, Results & External Referral	High	Release 2	US-12.1, US-12.2, US-12.3, US-12.4
EPIC-13	Emergency, Diagnostic & Referral Services	Daily Diagnostic Activity Summary	Medium	Release 2	US-13.1
EPIC-14	Pharmacy, Inventory & Laboratory Management	Medicine Availability and Prescription Dispensing	High	Release 1	US-14.1, US-14.2, US-14.3
EPIC-15	Pharmacy, Inventory & Laboratory Management	Inventory Stock and Expiry Management	High	Release 1	US-15.1, US-15.2, US-15.3
EPIC-16	Pharmacy, Inventory & Laboratory Management	Procurement and Inventory Reporting	High	Release 2	US-16.1, US-16.2
EPIC-17	Pharmacy, Inventory & Laboratory Management	Laboratory Request and Result Processing	High	Release 2	US-17.1, US-17.2, US-17.3
EPIC-18	Financial, Billing & Reimbursement Management	Medical Expense Reimbursement Management	High	Release 1	US-18.1, US-18.2, US-18.3
EPIC-19	Financial, Billing & Reimbursement Management	Referral Billing Management	High	Release 1	US-19.1, US-19.2
EPIC-20	Financial, Billing & Reimbursement Management	Accounts & Payment Processing	High	Release 2	US-20.1, US-20.2, US-20.3
EPIC-21	Financial, Billing & Reimbursement Management	Procurement Management	High	Release 2	US-21.1, US-21.2, US-21.3, US-21.4
EPIC-22	Administrative, Reporting & Scheduling System	Duty Roster Planning and Publication	High	Release 1	US-22.1, US-22.2, US-22.3

Epic ID	Subdomain	Epic Name	Priority	Release	Story IDs
EPIC-23	Administrative, Reporting & Scheduling System	Duty Changes, Absence and Cover	High	Release 1	US-23.1, US-23.2, US-23.3
EPIC-24	Administrative, Reporting & Scheduling System	Schedule Visibility	High	Release 1	US-24.1, US-24.2
EPIC-25	Administrative, Reporting & Scheduling System	Night On-Call Operations	High	Release 1	US-25.1, US-25.2
EPIC-26	Administrative, Reporting & Scheduling System	Laboratory Service-Count Reporting	Medium	Release 2	US-26.1, US-26.2
EPIC-27	Administrative, Reporting & Scheduling System	Policy Adherence Audit	High	Release 3	US-27.1, US-27.2

Common Acceptance Criteria

AC-01–AC-08 are shared criteria that apply across multiple stories. Each criterion keeps Given, When, and Then together in one cell, with its ID and applicability shown separately.

AC ID	Applies To	Acceptance Criteria
AC-01	All protected user stories.	Given: a protected function When: a user attempts access Then: the function and data are available only when the user is authenticated and authorized for the required role.
AC-02	All data-entry and decision stories.	Given: required input for a workflow When: the user submits incomplete or invalid data Then: the system blocks only the invalid operation, identifies the problem clearly, and preserves valid previously entered data where practical.
AC-03	All record-changing stories.	Given: a successful create/update action When: the transaction completes Then: the resulting record is stored against the correct patient, case, transaction, schedule, report, or other domain entity.
AC-04	Stories with external/subsystem dependencies.	Given: a required external or subsystem dependency is unavailable When: a dependent operation is attempted Then: the system reports the dependency failure and does not silently create inconsistent or partial final data.

AC ID	Applies To	Acceptance Criteria
AC-05	Audit-relevant clinical, emergency, financial, inventory, scheduling and reporting stories.	Given: an audit-relevant action When: it completes or is rejected Then: the responsible actor and timestamp are retained with the material status/decision where the source requires traceability.
AC-06	Stories handling sensitive data.	Given: protected medical, diagnostic, referral, financial, or identity data When: information is displayed, exported, logged, or notified Then: only information permitted for the requesting role/channel is exposed.
AC-07	Stories that finalize or lock records.	Given: a final, verified, approved, submitted, or otherwise locked record When: a later correction is required Then: the original record is not silently overwritten and the approved amendment/version mechanism is used.
AC-08	Transactional and integration stories.	Given: an operation fails before completion When: failure handling finishes Then: the system remains in a consistent state and the user receives an actionable status or retry path where the source requires one.

4.2 Patient Care & Medical Record Management

This subsection presents the Patient Care & Medical Record Management backlog within the global system numbering scheme. All stories use the common fields, priority scale, and acceptance-criteria format.

4.2.1 EPIC-1: Authentication and Session Management

Epic Description and Priority

Epic Field	Description
Description	Authenticates users whose institutional accounts are provisioned by the IUT ICT Center, enforces first-time password change where required, supports password maintenance, and securely ends user sessions. Institutional user registration/account creation is outside this system.
Business Value	Provides controlled access to Medical Center functions without duplicating the IUT ICT Center's account-provisioning responsibility.
Priority	High
Target Release	Release 1

User Stories

US-1.1: Log In	
ID	US-1.1
Epic	EPIC-1: Authentication and Session Management
Story	As an ICT-provisioned user, I want to log in with my registered IUT email and password, so that I can access the functions permitted for my role.
Priority	High
Precondition	• The IUT ICT Center has provisioned an active institutional account and required authorization information. • The user has valid login credentials. • The authentication service is operational.
Postcondition	• The user is authenticated and a secure session is established. • The system proceeds to first-time password change or automatically loads the dashboard/workspace for the user's authorized production role according to ICT-provided authorization data. • The production login flow does not ask the user to choose a role; any role-selection control shown in the wireframe exists only for presentation/demo purposes.
Story Points	
Dependencies	IUT ICT Center Identity and Account Provisioning; Authentication Service; Audit Logging Service
Definition of Done	• Login flow, invalid-credential handling and role handoff are tested. • Integration with ICT-provided account status is verified.

Story ID	AC ID	Acceptance Criteria
US-1.1	AC-09 [Derived]	Given: an active ICT-provisioned account and valid credentials When: the user logs in Then: the user is authenticated, a secure session is established, and the correct authorized role dashboard loads automatically (or first-time password change is shown when required), without a production role-selection control.

US-1.2: Complete First-Time Login	
ID	US-1.2
Epic	EPIC-1: Authentication and Session Management

Story	As a newly provisioned user, I want the system to require a password change during my first login when the account is flagged for first-time access, so that temporary credentials cannot be used for normal access.
Priority	High
Precondition	- The user has authenticated successfully. - The ICT-provided account status is marked First Time Login.
Postcondition	- The user is redirected to Change Password. - Protected application access remains restricted until the password is changed successfully.
Story Points	
Dependencies	US-1.1 Log In; US-1.3 Change Password; ICT-provided First Time Login status
Definition of Done	First-time access cannot bypass the password-change requirement.

Story ID	AC ID	Acceptance Criteria
US-1.2	AC-10 [Derived]	Given: the user has authenticated successfully and the account is marked First Time Login When: first-time access continues Then: the user is redirected to Change Password and protected access remains restricted until the password change succeeds.

US-1.3: Change Password	
ID	US-1.3
Epic	EPIC-1: Authentication and Session Management
Story	As an authenticated user, I want to replace my temporary or current password with a valid new password, so that my account remains secure.
Priority	High
Precondition	- The user is authenticated. - The user can access the Change Password function.
Postcondition	- The password is securely updated and the previous password becomes invalid. - The First Time Login restriction is cleared when applicable.
Story Points	
Dependencies	US-1.1 Log In; Password Policy; Authentication Service; Audit Logging Service

Definition of Done	• Password-policy validation and previous-password invalidation are tested.
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Story ID	AC ID	Acceptance Criteria
US-1.3	AC-11 [Derived]	Given: an authenticated user on Change Password When: a valid new password is submitted Then: the password is securely updated, the previous password becomes invalid, and the First Time Login restriction is cleared when applicable.

US-1.4: Log Out	
ID	US-1.4
Epic	EPIC-1: Authentication and Session Management
Story	As an authenticated user, I want to securely end my active session, so that another person cannot continue using my account.
Priority	High
Precondition	• The user has an active authenticated session.
Postcondition	• The active session and tokens are invalidated. • The user is redirected to Login and the logout event is recorded.
Story Points	
Dependencies	US-1.1 Log In; Session Management Service; Audit Logging Service
Definition of Done	• Session/token invalidation is verified and the logout event is traceable.

Story ID	AC ID	Acceptance Criteria
US-1.4	AC-12 [Derived]	Given: an authenticated user with an active session When: logout is completed Then: the session and tokens are invalidated, the user is returned to Login, and the logout event is recorded.

4.2.2 EPIC-2: Nurse and Medical Attendant Operations

Epic Description and Priority

Epic Field	Description
Description	Groups routine patient-care activity recording, prescription-based daily tracking, referral-proof recording, emergency procedure/dressing records, operational record/report review, submission to the CMO for routing, and monthly medicine-bill reporting. Routine vitals are recorded once and reused by emergency triage rather than duplicated.
Business Value	Creates one coherent operational workflow for the routine clinical and reporting responsibilities of nurses and medical attendants.
Priority	High
Target Release	Release 1 and Release 2

User Stories

US-2.1: Record Patient Care Activity	
ID	US-2.1
Epic	EPIC-2: Nurse and Medical Attendant Operations
Story	As a nurse or medical attendant, I want to record patient vitals, dressing procedures, or emergency procedures, so that completed care is preserved in the patient medical record.
Priority	High
Precondition	- The actor has an active registered account and is logged in. - The actor is authorized to record patient-care activities. - A valid patient record is available. - An active prescription may be viewed when available but is not mandatory for every activity.
Postcondition	- The selected care activity is saved under the correct patient. - The activity type, details, date, time, and responsible staff member are stored. - Authorized users can view the record according to role.
Story Points	
Dependencies	IUT ICT Center Identity and Account Provisioning; US-1.1 Log In; Patient and Medical Record Repository; Role-Based Access Control
Definition of Done	- Record Patient Care Activity is implemented and code reviewed. - Relevant unit/integration tests and acceptance criteria pass.

Story ID	AC ID	Acceptance Criteria
US-2.1	AC-13 [Derived]	Given: an authenticated and authorized nurse or medical attendant and a valid patient record When: a patient-care activity is submitted Then: the activity is saved under the correct patient with its type, details, date, time, and responsible staff member and is available only to authorized users.

US-2.2: Record Prescription for Daily Tracking	
ID	US-2.2
Epic	EPIC-2: Nurse and Medical Attendant Operations
Story	As a nurse or medical attendant, I want to place prescribed medicines, administration status, treatment follow-up, and related care instructions into daily tracking, so that day-to-day care can be monitored without duplicating the pharmacy dispensing transaction.
Priority	High
Precondition	- The actor is authenticated and authorized for daily care tracking. - A valid patient and an active prescription are available. - The prescription contains medicines, instructions, or follow-up activities that require daily tracking.
Postcondition	- A daily tracking entry is linked to the patient and source prescription. - Medicine-administration status, treatment follow-up, related care activity, timestamp, and responsible staff member can be recorded as applicable. - When medicine has been supplied, the tracking entry references the dispensing record from US-14.3; daily tracking itself does not deduct inventory.
Story Points	
Dependencies	US-1.1 Log In; US-3.2 Save Prescription; US-3.3 View Active Prescription; Patient Record; US-14.3 when a dispensing record is referenced
Definition of Done	- Record Prescription for Daily Tracking is implemented and code reviewed. - Relevant unit/integration tests and acceptance criteria pass.

Story ID	AC ID	Acceptance Criteria
US-2.2	AC-14 [Derived]	Given: an active prescription containing items that require daily follow-up When: authorized staff add them to daily tracking Then: the tracking entry is linked to the patient and prescription and records the applicable administration, treatment, or care status; any actual medicine issue and inventory deduction remain part of US-14.3.

US-2.3: Record Submitted Referral Proof	
ID	US-2.3
Epic	EPIC-2: Nurse and Medical Attendant Operations
Story	As a nurse or medical attendant, I want to record referral proof submitted by a patient or dependent, so that evidence of external medical care is preserved.
Priority	Medium
Precondition	• The actor has an active account and is logged in. • A valid patient or dependent exists. • Referral proof has been submitted. • The proof is readable enough to record.
Postcondition	• The proof is stored under the correct patient or dependent. • Submission details and document references are stored. • Date, time, and receiving staff identity are recorded. • Authorized staff can access the proof for follow-up.
Story Points	
Dependencies	US-1.1 Log In; Patient Record; Referral Record; Secure Document Storage
Definition of Done	• Record Submitted Referral Proof is implemented and code reviewed. • Relevant unit/integration tests and acceptance criteria pass.

Story ID	AC ID	Acceptance Criteria
US-2.3	AC-15 [Derived]	Given: an authenticated and authorized nurse or medical attendant and readable referral proof When: the proof is recorded Then: it is stored under the correct patient or dependent with submission details, document reference, timestamp, and receiving staff identity and is available to authorized staff for follow-up.

US-2.4: Review and Submit Operational Records / Reports to CMO	
ID	US-2.4
Epic	EPIC-2: Nurse and Medical Attendant Operations

Story	As a nurse or medical attendant, I want to review the operational records/reports I am permitted to access and submit an updated record or report to the CMO, so that the CMO can review it and route it to the appropriate department without creating duplicate records.
Priority	Medium
Precondition	• The actor has an active account and is authorized. • The relevant record/report exists and is within the actor's permission scope. • Permitted record/report types include Daily Medicine Stock, Monthly Medicine Bill, Emergency Procedure Record, Dressing Procedure Record, Referral Record, Prescription Record, and published Lab Reports.
Postcondition	• The selected updated record/report is submitted to the CMO with its source reference preserved. • Submission date, time, preparer identity, and status are recorded. • The submitted version is read-only to the submitting actor except through a controlled new version/amendment. • The CMO can review the item, choose an appropriate destination department, forward it, and the routing status/history is retained.
Story Points	
Dependencies	US-2.1 Record Patient Care Activity; US-2.2 Daily Tracking; US-2.3 Referral Proof; US-2.5 Monthly Medicine Bill; US-3.3 Prescription View; US-17.3 Published Lab Reports; Medicine/Inventory Records; CMO Records Inbox / Routing
Definition of Done	• Authorized record/report viewing, CMO submission, and source-reference preservation are implemented. • CMO review, destination selection, forwarding, and routing-status history are tested. • Relevant unit/integration tests and acceptance criteria pass.

Story ID	AC ID	Acceptance Criteria
US-2.4	AC-16 [Derived]	Given: an authorized nurse or medical attendant reviewing a permitted operational record/report When: the actor submits an updated record/report to the CMO Then: the source reference, submitter, timestamp, and status are preserved; the CMO can review and forward it to an appropriate department; and the routing history remains traceable.

US-2.5: Prepare Monthly Medicine Bill Report	
ID	US-2.5
Epic	EPIC-2: Nurse and Medical Attendant Operations

Story	As a nurse or medical attendant, I want to prepare a monthly medicine-bill report, so that medicine transactions and costs are summarized for later financial review.
Priority	Medium
Precondition	• The actor has an active account and is authorized to prepare reports. • A reporting month and year are selected. • Relevant dispensing and cost records exist, or a zero-activity report is permitted.
Postcondition	• The monthly report is generated and stored. • The period, totals, date, and preparer identity are recorded. • The report is marked Draft or Prepared.
Story Points	
Dependencies	US-1.1 Log In; Medicine Dispensing Records; Medicine Cost Records; Export Service; Audit Logging Service
Definition of Done	• Prepare Monthly Medicine Bill Report is implemented and code reviewed. • Relevant unit/integration tests and acceptance criteria pass.

Story ID	AC ID	Acceptance Criteria
US-2.5	AC-17 [Derived]	Given: an authorized nurse or medical attendant and a selected reporting month When: the monthly medicine-bill report is prepared Then: the report is generated and stored with the reporting period, totals, date, preparer identity, and Draft or Prepared status.

4.2.3 EPIC-3: Prescription Management

Epic Description and Priority

Epic Field	Description
Description	Supports creation, finalization, active viewing, and historical viewing of patient prescriptions while preserving the required information and presentation baseline of the pre-existing prescription format. The prescription record is structured as one document rather than scattered editable panels.
Business Value	Creates structured, permanent, accessible, and read-only prescription documents that support continuity of treatment, nursing follow-up, and downstream dispensing.
Priority	High
Target Release	Release 1

Required Prescription Information

- Every finalized prescription shall contain the following required information: Full Name of Patient; Designation / ID No.; Mobile No.; Nationality; Date; Age; Sex; Vitals; Chief Complaints and Duration; Investigation; Advice / Instructions; Prescribed Medications; and Sign of Treating Physician.
- Patient identity fields sourced from the authoritative institutional/patient record shall be displayed as read-only unless a controlled correction process applies.
- A separate standalone Diagnosis panel is not required by the wireframe. Clinical assessment and any diagnostic findings remain part of the clinical record, while laboratory/diagnostic findings are presented in the consolidated Lab & Diagnostic Results area.

User Stories

US-3.1: Create New Prescription	
ID	US-3.1
Epic	EPIC-3: Prescription Management
Story	As a doctor, I want to create a prescription draft for a valid patient using the approved prescription fields, so that the patient's complaints, vitals, investigations, advice, and medicines are recorded in one structured document.
Priority	High
Precondition	• The doctor has an active account and is logged in. • The patient exists in the authoritative institutional/patient record. • Before the consultation starts, the doctor can view the available prior patient context: profile, allergies/alerts, latest vitals, previous consultations, prescriptions, referrals, emergency history, and published laboratory/diagnostic results.
Postcondition	• A prescription draft is created with a unique prescription ID under the patient. • The draft supports all required prescription information defined above. • Treating-physician identity is linked to the draft and the Sign of Treating Physician field is required before finalization.
Story Points	
Dependencies	US-1.1 Log In; IUT ICT Center Identity / Account Interface; Unified Patient Record; Published Lab & Diagnostic Results
Definition of Done	• Required prescription fields and validation are implemented. • Pre-consultation prior-record context is available to the doctor without duplicating source records. • Relevant unit/integration tests and acceptance criteria pass.

Story ID	AC ID	Acceptance Criteria
US-3.1	AC-18 [Derived]	Given: an authenticated doctor and a valid patient record When: the doctor creates a new prescription Then: a draft with a unique prescription ID is created and supports the required patient identity, vitals, chief complaints/duration, investigation, advice/instructions, prescribed medications, and Sign of Treating Physician field.

US-3.2: Finalize Prescription	
ID	US-3.2
Epic	EPIC-3: Prescription Management
Story	As a doctor, I want to finalize a complete prescription draft, so that it becomes an active, locked prescription document available to authorized users.
Priority	High
Precondition	- The doctor has an active account and is logged in. - A prescription draft exists and every required field is complete. - The Sign of Treating Physician is present.
Postcondition	- The prescription status becomes Active. - Standard editing is locked; later correction uses a controlled amendment/version rather than destructive editing. - The final document becomes available to the patient and authorized medical staff and may be used for dispensing.
Story Points	
Dependencies	US-3.1 Create New Prescription; Prescription Repository; Role-Based Access Control
Definition of Done	- Final prescription output is checked against the approved prescription format and required information list. - Finalization, locking, and controlled amendment behavior are tested. - Relevant unit/integration tests and acceptance criteria pass.

Story ID	AC ID	Acceptance Criteria
US-3.2	AC-19 [Derived]	Given: an authenticated doctor and a complete prescription draft When: the doctor finalizes the prescription Then: its status becomes Active, standard editing is locked, and the final prescription document becomes available to the patient and authorized medical staff.

US-3.3: View Active Prescription	
ID	US-3.3
Epic	EPIC-3: Prescription Management
Story	As an authorized doctor, patient, nurse, or medical attendant, I want to view an active prescription document, so that the prescribed medicines, investigations, advice, and instructions can be followed without modifying the source record.
Priority	High
Precondition	- The user is authenticated with permission to view the target patient. - At least one active prescription exists.
Postcondition	The active prescription document is displayed accurately and read-only.
Story Points	
Dependencies	US-3.2 Finalize Prescription; Role-Based Access Control; Patient Record
Definition of Done	- Active prescription view is implemented for each authorized role and preserves the approved document structure. - Relevant unit/integration tests and acceptance criteria pass.

Story ID	AC ID	Acceptance Criteria
US-3.3	AC-20 [Derived]	Given: an authenticated user with permission to view the target patient When: the active prescription is opened Then: the current active prescription document is displayed accurately without modifying it.

US-3.4: View Prescription History	
ID	US-3.4
Epic	EPIC-3: Prescription Management
Story	As an authorized doctor, patient, nurse, or medical attendant, I want to view past prescriptions chronologically, so that previous treatment and instructions can support continuity of care.
Priority	Medium
Precondition	- The user is authenticated and authorized. - The patient has an existing medical record.

Postcondition	A chronological list of permitted past prescription documents is displayed read-only.
Story Points	
Dependencies	US-3.2 Finalize Prescription; Patient Medical Record Repository; Role-Based Access Control
Definition of Done	- Prescription history is implemented and respects role-based access. - Relevant unit/integration tests and acceptance criteria pass.

Story ID	AC ID	Acceptance Criteria
US-3.4	AC-21 [Derived]	Given: an authenticated user with permission to view prescription history When: prescription history is requested Then: a chronological list of permitted past prescription documents is displayed without modifying the records.

4.2.4 EPIC-4: Laboratory Test Management

Epic Description and Priority

Epic Field	Description
Description	Enables doctors to create complete laboratory-test requests, set priority, route requests to the laboratory queue, and cancel eligible pending requests.
Business Value	Ensures diagnostic requests are correctly linked, prioritized, and traceable.
Priority	High
Target Release	Release 2

User Stories

US-4.1: Request Laboratory Test	
ID	US-4.1
Epic	EPIC-4: Laboratory Test Management
Story	As a doctor, I want to create and prioritize a laboratory-test request for a patient, so that the Lab Technician receives a complete and traceable diagnostic request.
Priority	High

Precondition	- The doctor has an active account and is logged in. - The patient has a valid ICT Center record. - The system and laboratory queue are available.
Postcondition	- A unique laboratory request is created and linked to the patient. - Status is set to Requested. - The request appears in the Lab Technician pending queue. - The request is added to laboratory history.
Story Points	
Dependencies	US-1.1 Log In; IUT ICT Center Identity / Account Interface; Patient Record; Laboratory Queue
Definition of Done	- Request Laboratory Test is implemented and code reviewed. - Relevant unit/integration tests and acceptance criteria pass.

Story ID	AC ID	Acceptance Criteria
US-4.1	AC-22 [Derived]	Given: an authenticated doctor and a valid patient record When: a laboratory test request is submitted Then: a unique request is linked to the patient with status Requested, appears in the laboratory pending queue, and is added to the patient laboratory history.

4.2.5 EPIC-5: Emergency Examination Services

Epic Description and Priority

Epic Field	Description
Description	Supports medical-certificate issuance, emergency-examination request submission, verification, decision, and scheduling.
Business Value	Provides a fair and medically verified process for students requiring examination-related accommodation or an emergency examination.
Priority	High
Target Release	Release 2

User Stories

US-5.1: Obtain Medical Certificate

ID	US-5.1
Epic	EPIC-5: Emergency Examination Services
Story	As a student, I want to request and obtain a medically justified certificate, so that verified medical evidence can support an examination-related request.
Priority	High
Precondition	• The student has a condition that may affect examination attendance or performance. • The student can provide valid identification. • Authorized medical personnel are available. • Required medical information and supporting evidence are available when applicable.
Postcondition	• The condition is assessed. • A certificate is issued when the condition is confirmed. • The certificate includes student details, assessment, issue date, recommendation, signature, and authorization details. • The certificate becomes available for an emergency-examination request. • If not verified, no certificate is issued and the reason is recorded.
Story Points	
Dependencies	US-1.1 Log In; Valid Student Record; Authorized Medical Personnel; Secure Certificate Storage
Definition of Done	• Obtain Medical Certificate is implemented and code reviewed. • Relevant unit/integration tests and acceptance criteria pass.

Story ID	AC ID	Acceptance Criteria
US-5.1	AC-23 [Derived]	Given: a student with a condition that may affect examination attendance or performance When: the medical assessment is completed Then: a certificate is issued only when the condition is confirmed, includes the required student, assessment, date, recommendation, signature, and authorization details, and becomes available for an emergency-examination request; otherwise the reason for non-issuance is recorded.

US-5.2: Request Emergency Examination	
ID	US-5.2
Epic	EPIC-5: Emergency Examination Services

Story	As a student, I want to submit an emergency-examination request with a medical certificate and supporting documents, so that the Registrar Office can review my eligibility.
Priority	High
Precondition	• The student has an active account and is logged in. • The student is enrolled in the relevant course or examination. • A valid medical reason exists. • A valid medical certificate has been obtained. • Supporting documents are available.
Postcondition	• The request and documents are stored. • The request is forwarded to the Registrar Office. • The student receives confirmation. • No examination is scheduled before approval.
Story Points	
Dependencies	US-1.1 Log In; US-5.1 Obtain Medical Certificate; Registrar Examination Records; Secure Document Storage
Definition of Done	• Request Emergency Examination is implemented and code reviewed. • Relevant unit/integration tests and acceptance criteria pass.

Story ID	AC ID	Acceptance Criteria
US-5.2	AC-24 [Derived]	Given: an authenticated student with the required supporting documents When: an emergency-examination request is submitted Then: the request and documents are stored, forwarded to the Registrar Office, and acknowledged to the student, and no examination is scheduled before approval.

US-5.3: Approve Emergency Examination Request	
ID	US-5.3
Epic	EPIC-5: Emergency Examination Services
Story	As a Registrar Office user, I want to verify and decide an emergency-examination request and schedule it when approved, so that the outcome is fair, traceable, and medically supported.
Priority	High

Precondition	• The request has been submitted. • The certificate and supporting documents are available. • The Registrar Office user is authorized. • The certificate can be verified. • No final decision already exists.
Postcondition	• The request becomes Approved, Rejected, or Clarification Required. • Certificate verification status is updated. • Approved requests are scheduled with venue, date, time, and invigilator. • Medical Center arrangements are recorded when needed. • Rejection reasons are recorded and communicated.
Story Points	
Dependencies	US-5.2 Request Emergency Examination; IUT Medical Center Verification; Examination Scheduling; Invigilator Availability; Notification Service
Definition of Done	• Approve Emergency Examination Request is implemented and code reviewed. • Relevant unit/integration tests and acceptance criteria pass.

Story ID	AC ID	Acceptance Criteria
US-5.3	AC-25 [Derived]	Given: a submitted emergency-examination request When: an authorized Registrar Office user reviews it Then: the request receives an Approved, Rejected, or Clarification Required status; certificate verification is updated; approved requests are scheduled with venue, date, time, and invigilator; and rejection reasons are recorded and communicated.

4.3 Emergency, Diagnostic & Referral Services

This subsection covers emergency registration/triage, ambulance and observation workflows, referral processing, diagnostic booking/execution, results and diagnostic reporting.

4.3.1 EPIC-6: Emergency Registration & Identity Verification

Epic Description and Priority

Epic Field	Description
Description	Enables a patient (or accompanying peer, if the patient is unresponsive) to be registered as an emergency case within minutes of arrival, with identity verified against the university ID database where possible. This is the entry point to every downstream Emergency, Diagnostic & Referral Services workflow.

Epic Field	Description
Business Value	Removes the current paper-based registration bottleneck and ensures every emergency encounter has a traceable Case ID from the first moment of contact, without ever delaying treatment for the sake of administrative completeness.
Priority	High
Target Release	Release 1

User Stories

US-6.1: Register Emergency Patient	
ID	US-6.1
Epic	EPIC-6: Emergency Registration & Identity Verification
Story	As a nurse or medical attendant, I want to register an arriving emergency patient and open a case, so that treatment can begin without delay.
Priority	High
Precondition	• Patient or an accompanying person has arrived at the Emergency desk and is a registered member of the IUT community.
Postcondition	• A new emergency case record exists with a unique Case ID; the patient is queued for triage; the patient's existing medical profile (if available) is displayed.
Story Points	
Dependencies	US-6.2 (identity verification, included); US-6.3 (fallback path); Patient Care & Medical Record Management unified patient record API.
Definition of Done	• Code reviewed and merged; unit and integration tests passing. • Functional and NFR Acceptance criteria verified. in staging. • Triage-tablet UI reviewed for one-handed, high-contrast, large-touch-target usability. • Patient Care & Medical Record Management patient-record read/write contract confirmed with a live integration test. • Audit log entry generated and verified for every registration. • Quick-reference card updated and reviewed by a ward-staff representative.

Story ID	AC ID	Acceptance Criteria
US-6.1	AC-26	Given: a patient (or peer) at the Emergency desk When: the nurse or medical attendant submits registration Then: a unique Case ID is generated and the patient is queued for triage.

Story ID	AC ID	Acceptance Criteria
US-6.1	AC-27	Given: a returning patient When: registration completes Then: the patient's existing medical profile (allergies, blood group, prior visits) is displayed if available.

US-6.2: Verify Patient Identity via ID Card	
ID	US-6.2
Epic	EPIC-6: Emergency Registration & Identity Verification
Story	As a nurse or medical attendant, I want the system to verify a patient's identity from their ID card, so that the correct patient category and history are linked to the case.
Priority	High
Precondition	• Patient's ID card or ID number is presented.
Postcondition	• Patient identity is confirmed and linked to the university record and patient category, or the case is flagged "Unverified – Guest" and registration continues.
Story Points	
Dependencies	University ID Database API (IUT ICT Center); included by US-6.1.
Definition of Done	• Code reviewed and merged; unit and integration tests passing. • ID-scanner hardware integration tested against the IUT ICT Center API in a staging environment. • Fallback manual ID-entry path tested end to end. • Acceptance criteria verified. • Audit trail of verification attempts (matched, unverified, manual) implemented and reviewed.

Story ID	AC ID	Acceptance Criteria
US-6.2	AC-28	Given: a scanned or manually entered ID number When: the system queries the university database Then: patient category, name, department, and medical-history flag are returned.
US-6.2	AC-29	Given: no matching record is found When: verification is attempted Then: the patient is flagged "Unverified – Guest" and registration continues.

US-6.3: Register Patient via Peer (Fallback)
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ID	US-6.3
Epic	EPIC-6: Emergency Registration & Identity Verification
Story	As a nurse or medical attendant, I want to open a provisional emergency case using information supplied by an accompanying peer, so that an unresponsive patient's treatment is never delayed by identity verification.
Priority	High
Precondition	The primary patient is unresponsive, incapacitated, or unable to present ID.
Postcondition	A provisional case record marked "Pending Verification" (or "Unknown" if no information is available) exists and the patient proceeds directly to triage.
Story Points	
Dependencies	US-6.1 (extends); later verified-identity linkage depends on US-6.2.
Definition of Done	- Code reviewed and merged; unit and integration tests passing. "Pending Verification" cases visibly flagged on the admin dashboard for follow-up. - Verified-identity linkage workflow (linking a provisional record to a verified identity later) tested. - Acceptance criteria verified.

Story ID	AC ID	Acceptance Criteria
US-6.3	AC-30	Given: an unresponsive patient with an accompanying peer When: peer-supplied details are recorded Then: a provisional record marked "Pending Verification" is created and the case proceeds to triage without waiting for full verification.
US-6.3	AC-31	Given: no identifying information is available at all When: registration is attempted Then: the patient is registered as "Unknown" with a temporary Case ID.

4.3.2 EPIC-7: Vital Signs, Triage & Ambulance Request / Live Tracking

Epic Description and Priority

Epic Field	Description
Description	Covers emergency triage using the patient's latest available vitals, the clinical decision to transfer, direct ambulance requests from the on-duty doctor or nurse to the on-duty ambulance driver, driver acceptance/rejection, and live map-based driver location/status visible to the requesting clinical staff after acceptance.

Epic Field	Description
Business Value	Keeps emergency triage concise, removes duplicate vital-sign entry, simplifies the ambulance request flow, and gives the clinical team direct visibility of the accepted ambulance during transfer.
Priority	High
Target Release	Release 1

User Stories

US-7.1: Reuse / Record Vitals and Assign Triage Level	
ID	US-7.1
Epic	EPIC-7: Vital Signs, Triage & Ambulance Request / Live Tracking
Story	As a nurse or medical attendant, I want emergency triage to reuse the patient's latest valid routine vitals and capture only triage-specific assessment data, so that the emergency record is clear and does not duplicate the vitals record.
Priority	High
Precondition	- The patient has been registered fully or provisionally under US-6.1/US-6.3. - If current valid vitals already exist, they are available for linking; if no usable reading exists, authorized staff may record a new vital-sign record before or during triage.
Postcondition	- The emergency case links to the relevant vital-sign record rather than copying the values into a second editable record. - Triage stores severity (Critical/Urgent/Non-urgent), consciousness/immediate-risk assessment, presenting emergency, initial assessment, recommended next action, timestamp, and responsible staff. - The next action may be immediate doctor review, emergency procedure, Bed Rest / Observation, or ambulance preparation/request.
Story Points	
Dependencies	US-6.1/US-6.3; US-2.1 routine vitals; feeds US-7.2 ambulance request and US-8.1 Bed Rest / Observation.
Definition of Done	- Existing-vitals linkage and no-duplicate-entry behavior are tested. - Numeric range validation is tested when a new vital-sign record is required. - Critical triage bypasses the standard queue and shows the appropriate immediate next action. - Doctor review of nurse-assigned triage is supported where required.

Story ID	AC ID	Acceptance Criteria
US-7.1	AC-32	Given: a registered emergency patient with a latest valid vital-sign record When: the nurse opens emergency triage Then: the existing vitals are linked/read-only and the nurse records only the triage-specific severity and assessment data.
US-7.1	AC-33	Given: a critical triage level When: it is assigned Then: the case bypasses the standard queue and the system recommends immediate doctor attention, an emergency procedure, or ambulance preparation as appropriate.

US-7.2: Call Ambulance from Emergency Care	
ID	US-7.2
Epic	EPIC-7: Vital Signs, Triage & Ambulance Request / Live Tracking
Story	As an on-duty doctor or nurse, I want to call an ambulance directly from the emergency case, so that the on-duty driver receives a complete request immediately.
Priority	High
Precondition	- An emergency or night on-call case exists and transfer/ambulance assistance is clinically required. - The requester is an authorized on-duty doctor or nurse. - Destination, priority, reason, patient/case reference, and handover note are available.
Postcondition	- A unique ambulance request is stored against the Case ID with Requested By, destination, priority, reason, and handover note. - The request is delivered directly to the eligible on-duty ambulance driver. - Request status becomes Awaiting Driver until the driver accepts or rejects. - If no eligible driver is available, the requesting clinical team is shown the exception immediately and may record an approved alternate-transport arrangement.
Story Points	
Dependencies	US-7.1; US-25.2 for night on-call handoff; Transport Office driver-duty schedule (read-only); approved hospital/destination data.
Definition of Done	- Doctor/nurse ambulance request form and direct driver queue are tested. - Request source, Case ID, requester identity, destination, priority, and handover note remain traceable. - No duplicate ambulance request/transfer record is created. - No-driver and alternate-transport paths are tested.

Story ID	AC ID	Acceptance Criteria
US-7.2	AC-34	Given: an on-duty doctor or nurse has decided ambulance transport is required When: Call Ambulance is submitted Then: a request linked to the clinical Case ID is created and sent directly to the eligible on-duty driver with requester, destination, priority, reason, and handover information.
US-7.2	AC-35	Given: no eligible ambulance driver/vehicle is available When: the ambulance request is issued Then: the requesting doctor/nurse is shown the unavailability and may record an approved alternate-transport arrangement without deleting the original clinical transfer decision.

US-7.3: Driver Accepts Request and Shares Live Location	
ID	US-7.3
Epic	EPIC-7: Vital Signs, Triage & Ambulance Request / Live Tracking
Story	As the on-duty ambulance driver, I want to accept or reject an ambulance request and share my live trip status/location after acceptance, so that the requesting doctor or nurse can monitor where the ambulance is.
Priority	High
Precondition	- A valid ambulance request from US-7.2 is awaiting the driver. - The driver is authenticated and is the eligible/on-duty driver for the request.
Postcondition	- Accept records driver identity, vehicle, acceptance time, and active-trip status; Reject records the rejection/unavailability reason and returns the request status to the clinical team. - After acceptance, current location, last update time, ETA when available, and trip status are visible on a map-based panel to the requesting doctor and nurse. - The driver can update statuses such as Accepted, En route, Arrived at Pickup, Patient Onboard, Arrived at Hospital, and Completed. - On completion, the active request moves to read-only transfer history while remaining linked to the source Case ID.
Story Points	
Dependencies	US-7.2; approved driver mobile/GPS location source; map/location service; source emergency case.

Definition of Done	• Driver Accept/Reject is tested. • Live-location/status updates appear only after acceptance and are visible to authorized requesting clinical staff. • Completion produces a read-only transfer-history record linked to the source case. • Location-source failure does not delete the transfer; the latest known location/time and operational status remain visible with an appropriate warning.
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Story ID	AC ID	Acceptance Criteria
US-7.3	AC-36	Given: a valid ambulance request is awaiting the on-duty driver When: the driver accepts it Then: driver/vehicle/acceptance details are recorded and live map-based location/status becomes visible to the requesting doctor and nurse.
US-7.3	AC-37	Given: an accepted ambulance request When: the driver updates trip status/location or completes the transfer Then: authorized clinical staff see the latest location/status during the active trip and the completed transfer is retained read-only in history linked to the source Case ID.

4.3.3 EPIC-8: Bed Rest / Observation Management

Epic Description and Priority

Epic Field	Description
Description	Provides a separate Bed Rest / Observation workspace with exactly six beds. The bed board is independent from the emergency triage form and Emergency Procedure Record so that triage, procedures, and bed occupancy do not duplicate one another.
Business Value	Gives doctors and nurses a clear real-time six-bed availability view for on-site stabilization and prevents emergency triage/reporting fields from being mixed with bed management.
Priority	Medium
Target Release	Release 1

User Stories

US-8.1: Check and Reserve Bed Rest / Observation Bed	
ID	US-8.1

Epic	EPIC-8: Bed Rest / Observation Management
Story	As a doctor or nurse, I want to view the six Bed Rest / Observation beds and re-serve/update a bed separately from triage and procedure records, so that an on-site-treatable patient can be stabilized without ambiguity.
Priority	Medium
Precondition	A patient may require Bed Rest / Observation as indicated by the clinical/triage decision.
Postcondition	- The board shows all six beds and the real-time Free, Reserved, Occupied, or Cleaning status of each. - A reserved/occupied bed is linked to the Case ID/patient. - Discharge/transfer releases the bed through a controlled status update. - Bed changes do not modify the emergency triage or procedure record.
Story Points	
Dependencies	US-7.1; patient/case record.
Definition of Done	- Six-bed board is implemented and tested. - Reservation, occupancy, release, and logged discrepancy correction are tested. - Bed management remains separate from triage and emergency procedure data entry.

Story ID	AC ID	Acceptance Criteria
US-8.1	AC-38	Given: a patient needing Bed Rest / Observation When: authorized staff open the bed screen Then: exactly six beds are shown with their current statuses and an available bed can be reserved for the patient/case.
US-8.1	AC-39	Given: a discrepancy between system bed status and the physical bed state When: authorized staff correct it Then: the bed status is corrected and the discrepancy correction is logged without altering the source triage/procedure record.

4.3.4 EPIC-9: Referral Initiation & Documentation

Epic Description and Priority

Epic Field	Description
Description	Covers the doctor's decision to refer a patient externally, attaching a documented clinical recommendation, generating a formal referral letter, and – for external destinations – notifying the receiving hospital.
Business Value	Ensures every referral is policy-compliant (documented doctor's recommendation, authorized destination) and that receiving hospitals get timely, complete information.
Priority	High
Target Release	Release 1

User Stories

US-9.1: Initiate Patient Referral	
ID	US-9.1
Epic	EPIC-9: Referral Initiation & Documentation
Story	As a doctor, I want to open a referral case linked to the patient's record, so that a patient needing care beyond the Center's capability is formally referred.
Priority	High
Precondition	A patient's condition requires care beyond IUT Medical Center's capability or scope.
Postcondition	A referral case is opened with status "Initiated" and linked to the patient's Case ID.
Story Points	
Dependencies	US-9.2 (included); Patient Care & Medical Record Management patient record (contact info completeness).
Definition of Done	- Code reviewed and merged; unit and integration tests passing. - Prompt for missing required patient data (e.g./ contact info) tested. - Direct-initiation-from-ongoing-case path (emergency or diagnostics) tested. - Acceptance criteria verified.

Story ID	AC ID	Acceptance Criteria
US-9.1	AC-40	Given: a doctor decides external referral is necessary When: a referral is opened Then: it is created with status "Initiated" and linked to the patient's Case ID.

Story ID	AC ID	Acceptance Criteria
US-9.1	AC-41	Given: the patient's record is missing required data (e.g./ contact info) When: referral is attempted Then: the doctor is prompted to complete it first.

US-9.2: Attach Doctor's Recommendation	
ID	US-9.2
Epic	EPIC-9: Referral Initiation & Documentation
Story	As a doctor, I want to attach a written clinical recommendation to a referral, so that the receiving party has a documented reason and diagnosis summary.
Priority	High
Precondition	A referral has been initiated (US-9.1).
Postcondition	The recommendation is attached and version-locked once the referral moves past "Draft" status.
Story Points	
Dependencies	US-9.1 (includes this story); required before US-9.3.
Definition of Done	- Code reviewed and merged; unit and integration tests passing. - Rich-text recommendation editor tested for legibility on generated output. - Template-recommendation reuse path tested. - Acceptance criteria verified.

Story ID	AC ID	Acceptance Criteria
US-9.2	AC-42	Given: an initiated referral When: the doctor writes a recommendation Then: it is permanently attached to the referral record.

US-9.3: Generate Referral Letter	
ID	US-9.3
Epic	EPIC-9: Referral Initiation & Documentation
Story	As a doctor, I want to generate a formal referral letter compiling patient details, my recommendation, and relevant results, so that the receiving hospital or specialist has everything needed for handover.

Priority	High
Precondition	• Doctor's recommendation is attached (US-9.2) and, if applicable, upfront payment has been processed (US-10.1).
Postcondition	• A finalized referral letter, exportable as signed PDF, exists for the receiving hospital or specialist.
Story Points	
Dependencies	US-9.2; US-10.1; triggers US-9.4 for external destinations.
Definition of Done	• Code reviewed and merged; unit and integration tests passing. • PDF export format validated with a sample from each of the four partner hospitals' expected fields. • Manual-entry fallback for missing receiving-hospital details tested. • Acceptance criteria verified.

Story ID	AC ID	Acceptance Criteria
US-9.3	AC-43	Given: a recommendation is attached (and payment processed if applicable) When: the doctor finalizes the letter Then: it compiles patient details, recommendation, and relevant results into an exportable, signed PDF.

US-9.4: Notify Receiving Hospital	
ID	US-9.4
Epic	EPIC-9: Referral Initiation & Documentation
Story	As the system, I want to notify an external receiving hospital once a referral letter is finalized, so that the hospital is prepared for the incoming patient.
Priority	Medium
Precondition	• A referral letter has been finalized for an external hospital (US-9.3).
Postcondition	• The receiving hospital is notified via the configured channel, and confirmation of notification (or its absence) is logged.
Story Points	
Dependencies	US-9.3 (extends); external hospital notification channel.

Definition of Done	• Code reviewed and merged; unit and integration tests passing. • Delivery/acknowledgement logging implemented and tested for compliance. • Manual phone-follow-up logging path tested for unacknowledged notifications. • Acceptance criteria verified.
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Story ID	AC ID	Acceptance Criteria
US-9.4	AC-44	Given: a finalized referral letter for an external hospital When: notification is sent Then: the receiving hospital is informed via the configured channel and the event is logged.
US-9.4	AC-45	Given: an internal-network referral (not external) When: the letter is finalized Then: no external notification is triggered.

4.3.5 EPIC-10: Referral Payment & Refund Processing

Epic Description and Priority

Epic Field	Description
Description	Covers two linked but distinct money flows: an upfront referral counter payment received from the patient under the Pay-First policy, with receipt issuance, and a later refund that returns money against that original payment when the patient is found eligible. All monetary transactions are executed through the Financial, Billing & Reimbursement Management payment gateway; this epic covers the Emergency, Diagnostic & Referral Services workflow around initiating, tracking, and linking those transactions to the source referral.
Business Value	Keeps referral processing policy-compliant and auditable while integrating cleanly with the Financial, Billing & Reimbursement Management financial subsystem rather than duplicating billing logic.
Priority	High
Target Release	Release 1

User Stories

US-10.1: Process Upfront Referral Payment	
ID	US-10.1

Epic	EPIC-10: Referral Payment & Refund Processing
Story	As an Accounts / Finance / Cashier user, I want to collect and record the upfront referral fee, so that the referral can proceed per the Pay-First policy.
Priority	High
Precondition	Referral has been initiated (US-9.1) and requires payment under the Pay-First rule.
Postcondition	Payment is recorded against the Case ID via the Financial, Billing & Reimbursement Management gateway, and the referral case may proceed to letter generation (US-9.3).
Story Points	
Dependencies	US-9.1; Financial, Billing & Reimbursement Management payment gateway API; triggers US-10.2.
Definition of Done	- Code reviewed and merged; unit and integration tests passing. - Financial, Billing & Reimbursement Management payment-gateway integration tested for cash/card/mobile payment paths. - Deferred-payment (management-approved exception) path tested and logged. - PCI-relevant handling reviewed against Financial, Billing & Reimbursement Management's compliance checklist. - Acceptance criteria verified.

Story ID	AC ID	Acceptance Criteria
US-10.1	AC-46	Given: an initiated referral requiring payment When: the patient pays via the accounts counter or gateway Then: the payment is recorded against the Case ID and the referral can proceed.
US-10.1	AC-47	Given: a patient cannot pay upfront When: management approves an exception Then: deferred payment is logged as a recorded exception.

US-10.2: Issue Payment Receipt	
ID	US-10.2
Epic	EPIC-10: Referral Payment & Refund Processing
Story	As the system, I want to generate a receipt once a referral payment succeeds, so that the patient has documented proof of payment.
Priority	Medium
Precondition	A payment has been successfully processed (US-10.1).

Postcondition	• A sequentially numbered, tamper-evident receipt exists and has been provided to the patient (printed or digital).
Story Points	
Dependencies	Included by US-10.1.
Definition of Done	• Code reviewed and merged; unit and integration tests passing. • Digital-receipt fallback tested for printer-unavailable scenarios. • Duplicate-receipt-on-request path tested. • Acceptance criteria verified.

Story ID	AC ID	Acceptance Criteria
US-10.2	AC-48	Given: a successful payment When: the receipt is generated Then: it shows amount, date, service, and Case ID and is provided to the patient.

US-10.3: Process Refund	
ID	US-10.3
Epic	EPIC-10: Referral Payment & Refund Processing
Story	As an Accounts / Finance / Cashier user, I want to process a refund against an original referral payment, so that a patient later found eligible receives their money back.
Priority	Medium
Precondition	• An upfront payment was made for a referral (or diagnostic service) and the patient is later found eligible for a refund.
Postcondition	• The refund is processed through the Financial, Billing & Reimbursement Management gateway, the transaction ledger is updated, and a refund confirmation is issued.
Story Points	
Dependencies	US-10.1; Financial, Billing & Reimbursement Management payment gateway/refund API; Exceptional Refund Approval Authority (TBD) for disputed or over-limit cases.
Definition of Done	• Code reviewed and merged; unit and integration tests passing. • Traceability from refund to original payment record verified. • Escalation to the designated higher financial authority tested for disputed or over-limit refunds. • Acceptance criteria verified.

Story ID	AC ID	Acceptance Criteria
US-10.3	AC-49	Given: an approved refund-eligible payment When: the refund is processed Then: the transaction ledger is updated and a refund confirmation is issued.
US-10.3	AC-50	Given: a disputed or over-limit refund request When: additional authorization is required Then: the request is escalated to the designated higher financial authority for approval.

4.3.6 EPIC-11: Diagnostic Test Booking & Payment

Epic Description and Priority

Epic Field	Description
Description	Lets a patient or doctor check whether a diagnostic test can be performed on-site within the 8 AM–2 PM window, and collects the required upfront payment through the Financial, Billing & Reimbursement Management gateway before the test proceeds.
Business Value	Prevents wasted trips for tests the Center cannot perform, and enforces the Pay-First policy consistently for diagnostics.
Priority	High
Target Release	Release 2

User Stories

US-11.1: Check Internal Test Availability	
ID	US-11.1
Epic	EPIC-11: Diagnostic Test Booking & Payment
Story	As a Patient, I want to check whether a diagnostic test can be performed on-site, so that I know whether to expect an internal slot or an external-provider referral.
Priority	Medium
Precondition	• Patient wants to undergo a diagnostic test.
Postcondition	• Availability status (and, if available, an estimated slot time) is displayed; if unavailable, the flow proceeds to US-12.3.
Story Points	

Dependencies	Pharmacy, Inventory & Laboratory Management lab-equipment/technician scheduling API.
Definition of Done	• Code reviewed and merged; unit and integration tests passing. • Multi-test availability-check-in-one-visit path tested. • Operating-window (8 AM–2 PM) enforcement tested, including next-day queuing outside the window. • Acceptance criteria verified.

Story ID	AC ID	Acceptance Criteria
US-11.1	AC-51	Given: a requested test When: availability is checked Then: the system confirms whether it can be performed internally within the 8 AM–2 PM window and, if so, an estimated slot.
US-11.1	AC-52	Given: the requested test cannot be performed internally When: availability is checked Then: the flow proceeds to External Diagnostic Provider Recommendation (US-12.3).

US-11.2: Process Upfront Diagnostic Payment	
ID	US-11.2
Epic	EPIC-11: Diagnostic Test Booking & Payment
Story	As a Patient, I want to pay the test fee upfront through the payment gateway, so that my confirmed test slot is unlocked.
Priority	High
Precondition	• An internal test has been confirmed available and selected (US-11.1).
Postcondition	• Payment is confirmed and the “Perform Internal Test” step (US-12.1) is unlocked.
Story Points	
Dependencies	US-11.1; Financial, Billing & Reimbursement Management payment gateway API.

Definition of Done	• Code reviewed and merged; unit and integration tests passing. • Financial, Billing & Reimbursement Management payment-gateway integration tested (online and accounts-counter alternatives). • Payment-decline/retry path tested. • PCI-relevant handling reviewed against Financial, Billing & Reimbursement Management's compliance checklist. • Acceptance criteria verified.
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Story ID	AC ID	Acceptance Criteria
US-11.2	AC-53	Given: a confirmed, available internal test When: the patient completes payment Then: the payment is confirmed and test execution (US-12.1) is unlocked.
US-11.2	AC-54	Given: payment fails or is declined When: this occurs Then: the booking is held pending and the patient is notified to retry.

4.3.7 EPIC-12: Diagnostic Service Execution, Results & External Referral

Epic Description and Priority

Epic Field	Description
Description	Covers the diagnostic-service flow owned by Emergency, Diagnostic & Referral Services: performing a paid, confirmed on-site diagnostic service, recording its result, and generating an external-provider recommendation when that service cannot be completed on site. A case enters this flow when it is initiated through the diagnostic-service availability, booking, and payment path in EPIC-11. A doctor-issued formal laboratory request instead follows EPIC-4 and EPIC-17. Routing is determined by the initiating workflow, not by the test name alone.
Business Value	Closes the loop between test booking and clinically usable results, and ensures patients needing external testing are not left without direction.
Priority	High
Target Release	Release 2

User Stories

US-12.1: Perform Internal Diagnostic Test
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ID	US-12.1
Epic	EPIC-12: Diagnostic Service Execution, Results & External Referral
Story	As a doctor or nurse/medical attendant authorized for diagnostic services, I want to conduct a confirmed diagnostic service during operating hours, so that its result can be recorded and returned to the doctor.
Priority	High
Precondition	Payment has been processed and a test slot is confirmed within the 8 AM–2 PM operating window (US-11.2).
Postcondition	The test is conducted and results are ready for recording (US-12.2).
Story Points	
Dependencies	US-11.2; diagnostic-service availability/equipment schedule; triggers US-12.2.
Definition of Done	- Code reviewed and merged; unit and integration tests passing. - Technician-reassignment path tested for staff unavailability. - Redirect-to-external-provider path tested for equipment/expertise unavailability discovered at execution time. - Equipment calibration-log check step verified. - Acceptance criteria verified.

Story ID	AC ID	Acceptance Criteria
US-12.1	AC-55	Given: a confirmed, paid test slot When: the technician conducts the test within 8 AM–2 PM Then: Record Test Results (US-12.2) is triggered on completion.
US-12.1	AC-56	Given: the test proves not performable internally at execution time When: this is discovered Then: the flow redirects to External Diagnostic Provider Recommendation (US-12.3) instead of proceeding.

US-12.2: Record Test Results	
ID	US-12.2
Epic	EPIC-12: Diagnostic Service Execution, Results & External Referral
Story	As a doctor or nurse/medical attendant authorized for diagnostic services, I want to record the result of a completed diagnostic service and link it to the patient's Case ID, so that the treating doctor and permitted patient view can access it.

Priority	High
Precondition	A diagnostic test has been performed (US-12.1).
Postcondition	Results are stored, timestamped, technician-attributed, and linked to the Case ID; abnormal results are flagged for doctor review.
Story Points	
Dependencies	Included by US-12.1.
Definition of Done	- Code reviewed and merged; unit and integration tests passing. - Automated-equipment result-import path tested where applicable. - Abnormal/critical result auto-flagging tested. - Exportable, standardized result format validated for the permanent medical record. - Acceptance criteria verified.

Story ID	AC ID	Acceptance Criteria
US-12.2	AC-57	Given: a performed test When: the technician enters results Then: they are stored with timestamp and technician ID and become visible to the treating doctor and patient.
US-12.2	AC-58	Given: an abnormal/critical result When: recorded Then: the system automatically flags it for immediate doctor review.

US-12.3: Generate External Diagnostic Provider Recommendation	
ID	US-12.3
Epic	EPIC-12: Diagnostic Service Execution, Results & External Referral
Story	As a doctor or nurse/medical attendant authorized for diagnostic services, I want to generate an external-provider recommendation when the required diagnostic service cannot be completed on site, so that the patient can continue the diagnostic process externally.
Priority	Medium
Precondition	The requested test cannot be performed internally (equipment/expertise unavailable), per US-11.1 or US-12.1.
Postcondition	A recommendation document identifying the required diagnostic service and suggested approved external providers (US-12.4) is generated for the patient.
Story Points	

Dependencies	US-12.4 (included); triggered by US-11.1 or US-12.1.
Definition of Done	• Code reviewed and merged; unit and integration tests passing. • Patient-requested-specific-lab override path tested. • Manual-lookup fallback tested for uncovered test types. • Acceptance criteria verified.

Story ID	AC ID	Acceptance Criteria
US-12.3	AC-59	Given: internal testing is confirmed unavailable When: the recommendation is generated Then: it lists the required test and suggested approved providers (US-12.4).

US-12.4: Suggest External Diagnostic Provider Options	
ID	US-12.4
Epic	EPIC-12: Diagnostic Service Execution, Results & External Referral
Story	As the system, I want to match the required externally performed diagnostic service against the approved external-provider directory, so that the doctor can present suitable options to the patient.
Priority	Low
Precondition	• A recommendation for external testing is being generated (US-12.3).
Postcondition	• A ranked list of suitable approved external providers, including partner laboratories where applicable, is presented.
Story Points	
Dependencies	Included by US-12.3; administrator-maintained lab directory.
Definition of Done	• Code reviewed and merged; unit and integration tests passing. • Filter/sort by cost or distance tested. • Administrator directory-maintenance screen tested. • Acceptance criteria verified.

Story ID	AC ID	Acceptance Criteria
US-12.4	AC-60	Given: a required test type When: the directory is matched Then: a ranked list of suitable approved providers is returned.

Story ID	AC ID	Acceptance Criteria
US-12.4	AC-61	Given: no approved provider in the directory offers the required service When: matching fails Then: a generic search result or manual-lookup prompt is returned.

4.3.8 EPIC-13: Daily Diagnostic Activity Summary

Epic Description and Priority

Epic Field	Description
Description	Compiles a detailed clinical diagnostic-activity report for doctors and the CMO from the Emergency, Diagnostic & Referral Services workflow, including services performed, recorded result references, pending items, and external referrals. It is distinct from EPIC-26, which produces aggregate administrative counts and does not reproduce individual results.
Business Value	Gives doctors and the CMO a clinically useful end-of-day view of diagnostic activity while keeping aggregate administrative reporting separate.
Priority	Medium
Target Release	Release 2

User Stories

US-13.1: Generate Daily Clinical Diagnostic Activity Report	
ID	US-13.1
Epic	EPIC-13: Daily Diagnostic Activity Summary
Story	As a doctor, I want a detailed daily clinical diagnostic-activity report, so that performed services, result references, pending items, and external referrals are available for clinical record-keeping and CMO review.
Priority	Medium
Precondition	One or more diagnostic tests/activities have been recorded during the day.
Postcondition	A detailed clinical diagnostic-activity report is produced, exportable as PDF/Excel, and made available to authorized doctors and the CMO; aggregate administrative reporting remains within EPIC-26.
Story Points	

Dependencies	US-12.2, US-12.3; Unified Patient Record; Export Service.
Definition of Done	• Code reviewed and merged; unit and integration tests passing. • Custom-date-range generation path tested in addition to daily default. • “Pending” marking and later regeneration tested for incomplete same-day data. • Export format (PDF/Excel) validated for authorized clinical use and archival export. • Acceptance criteria verified.

Story ID	AC ID	Acceptance Criteria
US-13.1	AC-62	Given: recorded diagnostic activity for a day When: the report is generated Then: it compiles all tests performed, results recorded, and external-provider referrals for that period.
US-13.1	AC-63	Given: part of the day’s data is still pending When: the report is generated Then: those items are marked “Pending” and the report can be regenerated later.

4.4 Pharmacy, Inventory & Laboratory Management

This subsection covers medicine availability and dispensing, stock/expiry control, inventory reporting/procurement requests, and the formal laboratory request, processing, verification, and result-publication workflow. A doctor-issued laboratory request enters EPIC-4/EPIC-17 directly; it does not pass through the EPIC-11 payment/booking gate. EPIC-12 remains the separate diagnostic-service flow initiated through EPIC-11. Routing is determined by how the service is initiated, not by the test name alone.

4.4.1 EPIC-14: Medicine Availability and Prescription Dispensing

Epic Description and Priority

Epic Field	Description
Description	Enables medical staff to search medicine availability and stock levels, retrieve finalized prescriptions, and safely dispense prescribed medicines while ensuring automatic inventory deduction.
Business Value	Prevents dispensing out-of-stock medicines, eliminates manual record-keeping errors, and ensures accurate stock synchronization.
Priority	High
Target Release	Release 1

User Stories

US-14.1: Check Medicine Availability	
ID	US-14.1
Epic	EPIC-14: Medicine Availability and Prescription Dispensing
Story	As a doctor or nurse, I want to search for medicines and view their current stock levels, generic names, and dosage forms, so that I can prescribe or dispense available medicines.
Priority	High
Precondition	- The doctor or nurse is authenticated and authorized for this function. - Required dependencies are available: Authentication and Session Management; Inventory Database. - An authenticated doctor or nurse.
Postcondition	- The system displays current availability, remaining quantity, brands, and dosage forms. - The system displays “Requested Medicine Not Found”.
Story Points	
Dependencies	Authentication and Session Management; Inventory Database
Definition of Done	Medicine search field implemented; real-time inventory quantity and dosage forms displayed; out-of-stock messages shown when unavailable; acceptance criteria met.

Story ID	AC ID	Acceptance Criteria
US-14.1	AC-64	Given: an authenticated doctor or nurse When: a medicine name or generic name is entered Then: the system displays current availability, remaining quantity, brands, and dosage forms.
US-14.1	AC-65	Given: a requested medicine is not in the inventory When: searched Then: the system displays “Requested Medicine Not Found”.

US-14.2: Receive Prescription	
ID	US-14.2
Epic	EPIC-14: Medicine Availability and Prescription Dispensing
Story	As a nurse or medical attendant, I want to retrieve and review finalized prescriptions issued by doctors, so that I can prepare the correct medicines for dispensing.

Priority	High
Precondition	- The nurse or medical attendant is authenticated and authorized for this function. - Required dependencies are available: Patient Care & Medical Records Subsystem. - A nurse or medical attendant opens the pending prescriptions queue.
Postcondition	- The system displays the finalized prescription issued by the doctor. - The system displays an appropriate notification message.
Story Points	
Dependencies	Patient Care & Medical Records Subsystem
Definition of Done	Pending prescriptions queue functional; latest prescription details displayed clearly; access limited to authorized nurses; testing completed.

Story ID	AC ID	Acceptance Criteria
US-14.2	AC-66	Given: a nurse or medical attendant opens the pending prescriptions queue When: a patient is selected Then: the system displays the finalized prescription issued by the doctor.
US-14.2	AC-67	Given: no active prescription exists for the patient When: opened Then: the system displays an appropriate notification message.

US-14.3: Dispense Medicine	
ID	US-14.3
Epic	EPIC-14: Medicine Availability and Prescription Dispensing
Story	As a nurse or medical attendant, I want to record medicine dispensing against a valid prescription, so that patient medicines are handed out and inventory stock is automatically deducted.
Priority	High
Precondition	- The nurse or medical attendant is authenticated and authorized for this function. - Required dependencies are available: US-14.1, US-14.2. - A valid prescription and sufficient stock.
Postcondition	- The system issues the medicine and automatically updates inventory quantities. - The system dispenses available quantities, logs the shortfall, and updates stock.
Story Points	

Dependencies	US-14.1, US-14.2
Definition of Done	Dispensing action updates stock atomically; partial dispensing shortfall recorded; dispensing transaction saved and logged; tests pass.

Story ID	AC ID	Acceptance Criteria
US-14.3	AC-68	Given: a valid prescription and sufficient stock When: the nurse confirms dispensing Then: the system issues the medicine and automatically updates inventory quantities.
US-14.3	AC-69	Given: only partial stock is available When: dispensing is processed Then: the system dispenses available quantities, logs the shortfall, and updates stock.

4.4.2 EPIC-15: Inventory Stock and Expiry Management

Epic Description and Priority

Epic Field	Description
Description	Provides features to add, update, and remove medicine records, monitor stock quantities and expiration dates, and generate automated alerts for low stock or near-expiry items.
Business Value	Minimizes risk of medication shortages and prevents patient safety risks associated with expired drugs.
Priority	High
Target Release	Release 1

User Stories

US-15.1: Manage Inventory Records	
ID	US-15.1
Epic	EPIC-15: Inventory Stock and Expiry Management
Story	As a nurse or medical attendant with inventory permission, I want to add, update, or remove medicine records, so that the inventory database maintains accurate and up-to-date records.

Priority	High
Precondition	- The nurse or medical attendant is authenticated and has inventory permission. - Required dependencies are available: Inventory Database; Authentication and Session Management. - Authorized inventory permission is active.
Postcondition	- The system saves or updates the medicine record in the database. - The system blocks the action and displays a validation error. - The transaction is logged in the immutable audit log.
Story Points	
Dependencies	Inventory Database; Authentication and Session Management
Definition of Done	Inventory CRUD operations functional; input validation prevents duplicate IDs/invalid stock; all modifications logged in audit trail.

Story ID	AC ID	Acceptance Criteria
US-15.1	AC-70	Given: an authorized nurse or medical attendant When: valid medicine details are entered Then: the system saves or updates the medicine record in the database.
US-15.1	AC-71	Given: duplicate medicine identifiers or negative stock values When: submitted Then: the system blocks the action and displays a validation error.
US-15.1	AC-72	Given: any creation, modification, or deletion of inventory items When: completed Then: the transaction is logged in the immutable audit log.

US-15.2: Monitor Stock and Expiry Dates	
ID	US-15.2
Epic	EPIC-15: Inventory Stock and Expiry Management
Story	As a nurse or medical attendant with inventory permission, I want the system to continuously monitor stock levels and expiry dates, so that low-stock items and near-expiry medicines are easily identified.
Priority	High
Precondition	- The nurse or medical attendant is authenticated and has inventory permission. - Required dependencies are available: US-15.1. - Updated inventory records.

Postcondition	It compares current quantities against reorder thresholds and expiry dates against the system date.
Story Points	
Dependencies	US-15.1
Definition of Done	Automated monitoring logic operational; stock threshold and expiry date comparison functional; dashboard indicators updated.

Story ID	AC ID	Acceptance Criteria
US-15.2	AC-73	Given: updated inventory records When: the monitoring module runs Then: it compares current quantities against reorder thresholds and expiry dates against the system date.

US-15.3: Generate Low-Stock and Expiry Alerts	
ID	US-15.3
Epic	EPIC-15: Inventory Stock and Expiry Management
Story	As a nurse or medical attendant with inventory permission, I want to receive automated system alerts when medicine stock falls below reorder thresholds or approaches expiry, so that replenishment can be requested promptly.
Priority	High
Precondition	- The nurse or medical attendant is authenticated and has inventory permission. - Required dependencies are available: US-15.2. - Stock falls below predefined thresholds or expiry is within the warning window.
Postcondition	Prominent alert notifications are displayed.
Story Points	
Dependencies	US-15.2
Definition of Done	Automated alerts generated and highlighted on the dashboard; filter options for alert types implemented; tests pass.

Story ID	AC ID	Acceptance Criteria
US-15.3	AC-74	Given: stock falls below predefined thresholds or expiry is within the warning window When: the user logs in Then: prominent alert notifications are displayed.

4.4.3 EPIC-16: Procurement and Inventory Reporting

Epic Description and Priority

Epic Field	Description
Description	Enables the Chief Medical Officer (CMO) to view comprehensive inventory reports and formally submit procurement requests for stock replenishment.
Business Value	Streamlines restocking workflows, ensures administrative accountability, and supports procurement decision-making.
Priority	High
Target Release	Release 2

User Stories

US-16.1: View Inventory Reports	
ID	US-16.1
Epic	EPIC-16: Procurement and Inventory Reporting
Story	As the Chief Medical Officer, I want to view and filter up-to-date inventory reports, so that I can assess medicine availability, stock levels, and expiry statuses.
Priority	Medium
Precondition	- The Chief Medical Officer is authenticated and authorized for this function. - Required dependencies are available: EPIC-15. - An authenticated CMO.
Postcondition	- The system generates accurate reports reflecting real-time inventory records. - The report dynamically updates.
Story Points	
Dependencies	EPIC-15

Definition of Done	Report generation from latest database records implemented; filtering by date, category, and status functional; export/print capability verified.
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Story ID	AC ID	Acceptance Criteria
US-16.1	AC-75	Given: an authenticated CMO When: View Inventory Reports is selected Then: the system generates accurate reports reflecting real-time inventory records.
US-16.1	AC-76	Given: a generated report When: the CMO applies filters by date, medicine category, or stock status Then: the report dynamically updates.

US-16.2: Submit Procurement Request	
ID	US-16.2
Epic	EPIC-16: Procurement and Inventory Reporting
Story	As the Chief Medical Officer, I want to submit procurement requests for low-stock medicines to the Procurement Department, so that depleted stock is replenished in a timely manner.
Priority	High
Precondition	- The Chief Medical Officer is authenticated and authorized for this function. - Required dependencies are available: US-16.1, Financial, Billing & Reimbursement Management. - Identified low-stock medicines.
Postcondition	- The system creates a formal procurement request with the selected items and requested quantities. - The request stores the submission timestamp and CMO identity, receives status Pending Administrative Submission, and appears in the US-21.1 administrative procurement queue.
Story Points	
Dependencies	US-16.1; Financial, Billing & Reimbursement Management procurement interface
Definition of Done	Procurement selection and quantity adjustment form created; timestamp and submitting CMO identity logged; handoff to the US-21.1 administrative procurement queue verified.

Story ID	AC ID	Acceptance Criteria
US-16.2	AC-77	Given: identified low-stock medicines When: the CMO selects items, adjusts quantities, and confirms submission Then: the system creates a formal procurement request.
US-16.2	AC-78	Given: a submitted procurement request When: it is saved Then: it includes the submission timestamp and CMO identity, receives status Pending Administrative Submission, and appears in the US-21.1 administrative procurement queue.

4.4.4 EPIC-17: Laboratory Request and Result Processing

Epic Description and Priority

Epic Field	Description
Description	Manages the formal laboratory workflow from receiving doctor-issued laboratory requests to processing laboratory tests, verifying findings, and publishing final reports to patient medical records. This workflow is entered through EPIC-4 and does not use the EPIC-11 diagnostic-service booking/payment path.
Business Value	Eliminates paper-based lab reports, reduces reporting turnaround times, and ensures reliable diagnostic data integration.
Priority	High
Target Release	Release 2

User Stories

US-17.1: Receive Test Request	
ID	US-17.1
Epic	EPIC-17: Laboratory Request and Result Processing
Story	As a Laboratory Technician, I want to receive and review pending lab test requests submitted by doctors, so that I can verify patient details and prepare test equipment.
Priority	High

Precondition	- The Laboratory Technician is authenticated and authorized for this function. - Required dependencies are available: EPIC-4: Laboratory Test Management (Patient Care & Medical Record Management). - A doctor has submitted a lab request.
Postcondition	- The request appears in the pending queue. - The request status changes to “Received”.
Story Points	
Dependencies	EPIC-4: Laboratory Test Management (Patient Care & Medical Record Management)
Definition of Done	Pending lab request queue functional; patient identity matching verified; status updates to “Received”; testing completed.

Story ID	AC ID	Acceptance Criteria
US-17.1	AC-79	Given: a doctor has submitted a lab request When: the Laboratory Technician opens the lab portal Then: the request appears in the pending queue.
US-17.1	AC-80	Given: the Laboratory Technician verifies patient information and equipment readiness When: the Laboratory Technician accepts the request Then: the request status changes to “Received”.

US-17.2: Record and Process Laboratory Test	
ID	US-17.2
Epic	EPIC-17: Laboratory Request and Result Processing
Story	As a Laboratory Technician, I want to record laboratory test results into the system after performing tests, so that findings are digitally captured.
Priority	High
Precondition	- The Laboratory Technician is authenticated and authorized for this function. - Required dependencies are available: US-17.1. - A received lab request.
Postcondition	- The Laboratory Technician can enter and save the laboratory findings. - The system records the exception and alerts responsible personnel.
Story Points	
Dependencies	US-17.1

Definition of Done	Result entry interface operational; validation of complete laboratory-result fields; error/malfunction exception notification handled.
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Story ID	AC ID	Acceptance Criteria
US-17.2	AC-81	Given: a received lab request When: the test is conducted Then: the Laboratory Technician can enter and save the laboratory findings.
US-17.2	AC-82	Given: test failure due to equipment malfunction or invalid specimens When: logged Then: the system records the exception and alerts responsible personnel.

US-17.3: Verify and Publish Laboratory Results	
ID	US-17.3
Epic	EPIC-17: Laboratory Request and Result Processing
Story	As an authorized Laboratory Technician, I want to verify test result accuracy and publish a finalized laboratory report document, so that doctors, patients, nurses, and medical attendants with permission can view and download the final report.
Priority	High
Precondition	- The authorized user is authenticated and authorized for this function. - Required dependencies are available: US-17.2, Patient Care & Medical Record Management. - Recorded test results.
Postcondition	- The system publishes the finalized results to the Patient Care & Medical Records subsystem and generates/stores the corresponding final laboratory report document. - The report document is available for View Report and Download Document actions to authorized users. - Pending, Received, or Processing requests do not expose a final downloadable document. - The system blocks publication until formal verification is complete.
Story Points	
Dependencies	US-17.2, Patient Care & Medical Record Management

Definition of Done	• Result verification workflow complete; unverified results blocked from publication; secure transmission to Patient Care & Medical Record Management verified; audit log updated. • Final report-document preview and authorized download are tested; no final document is available before publication.
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Story ID	AC ID	Acceptance Criteria
US-17.3	AC-83	Given: recorded test results When: authorized Laboratory Technician complete verification and publish Then: the system publishes the finalized results, creates/stores the final laboratory report document, and makes View Report / Download Document available to authorized users.
US-17.3	AC-84	Given: an unverified or still-processing test result When: publication or final-document download is attempted Then: the system blocks final publication/download until formal verification and publication are complete.

4.5 Financial, Billing & Reimbursement Management

This subsection covers reimbursement, referral billing, accounts/payment processing, audit trails and procurement management.

4.5.1 EPIC-18: Medical Expense Reimbursement Management

Epic Description and Priority

Epic Field	Description
Description	Enables an eligible student, faculty member, or staff member to submit and track the institutional Request for Re-imbursement of Medical Expenses using the existing form structure, itemized supporting proofs, automatic total calculation, medical review by the CMO, and final Accounts payment processing.
Business Value	Digitizes the existing reimbursement form without losing its approved fields, reduces arithmetic/document errors, and keeps medical approval separate from payment execution.
Priority	High
Target Release	Release 1

Required Reimbursement Form Structure

- Claimant information shall include Name; Position / Student No. (or the applicable institutional identifier for faculty/staff); Bank A/C No.; Signature; and Dept. / Office.
- The expense table shall preserve four standard rows: Consultation fees; Medicines; Laboratory / Clinical Test; and Hospital / Clinic bills.
- Every expense row shall include Payment Document Serials, Taka amount, and its supporting proof/document upload.
- The system shall calculate the Total automatically from the entered Taka amounts; the claimant shall not type the grand total manually.
- The reimbursement form shall include an **Official Use Only** section. Fields such as Gross Payable Amount, CMO Decision, Approved Amount, and Accounts / Payment Status shall be system/reviewer controlled and shall not be editable by the claimant.

User Stories

US-18.1: Submit and Validate Medical Expense Claim	
ID	US-18.1
Epic	EPIC-18: Medical Expense Reimbursement Management
Story	As an eligible claimant, I want to submit the existing medical-expense reimbursement form with payment-document serials, amounts, and proof for each expense category, so that the system can calculate the total, validate the submission, and send it for CMO review.
Priority	High
Precondition	• The claimant is authenticated and eligible for reimbursement. • Claimant identity/category is available from the institutional record. • Required payment proofs and the relevant prescription/clinical reference are available where required by policy.
Postcondition	• The form stores claimant details, the four standard expense rows, payment-document serials, Taka values, and proof references. • The system automatically calculates and displays the Total. • Missing required proof, invalid reference, duplicate claim, or policy-invalid expense is identified with a specific reason. • A valid claim moves to Under CMO Review.
Story Points	
Dependencies	Institutional Identity / Eligibility; Patient Care & Medical Record Management prescription/clinical records; Secure Document Storage; reimbursement policy configuration

Definition of Done	- Existing-form field structure and the four standard expense categories are implemented. - Per-row proof upload, document-serial entry, and automatic Total calculation are tested. - Duplicate/reference/policy validation and clear rejection reasons are tested. - Successful submission moves to Under CMO Review.
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Story ID	AC ID	Acceptance Criteria
US-18.1	AC-85	Given: an eligible authenticated claimant with the required expense information and proofs When: the reimbursement claim is submitted Then: the system stores the four standard expense rows with document serials and proofs, automatically calculates the Total, records the claim as Submitted, and runs validation.
US-18.1	AC-86	Given: the system detects missing required proof, an invalid/duplicate claim, or another policy-invalid expense When: the claim is validated Then: the invalid submission is blocked or rejected with a specific reason while valid entered data is preserved where practical.

US-18.2: Track and View Claim Details	
ID	US-18.2
Epic	EPIC-18: Medical Expense Reimbursement Management
Story	As a claimant, I want to track my reimbursement status and open a claim to view its expenses, proof references, calculated/approved amounts, CMO decision, and Accounts payment status, so that the processing state is clear.
Priority	Medium
Precondition	- The claimant is authenticated and authorized. - At least one prior claim exists.
Postcondition	- A chronological claim list with totals and statuses is displayed. - Opening a claim shows its itemized expenses, proof references, CMO review state, approved amount when available, and Accounts/payment state.
Story Points	
Dependencies	US-18.1

Definition of Done	• Claim history and structured claim-detail view are implemented. • Status wording is consistent with the governed workflow vocabulary.
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Story ID	AC ID	Acceptance Criteria
US-18.2	AC-87	Given: a claimant has prior reimbursement claims When: Claim History or a claim detail is opened Then: the system shows a chronological list and the permitted itemized claim details, including calculated amount and current CMO/Accounts statuses.

US-18.3: Review and Route Reimbursement	
ID	US-18.3
Epic	EPIC-18: Medical Expense Reimbursement Management
Story	As the CMO, I want to review the itemized reimbursement claim alongside the relevant patient/prescription records and attached proofs, so that I can Accept or Reject it on medical/policy validity and route only accepted claims to Accounts.
Priority	High
Precondition	• The CMO is authenticated and authorized. • A validated claim from US-18.1 is awaiting CMO review. • Relevant claim proofs and patient/prescription records are available.
Postcondition	• Accept records the CMO decision/approved amount and routes the claim to the Accounts approved-payout queue. • Reject records a mandatory rejection reason and prevents payment processing. • The decision and timestamp are retained.
Story Points	
Dependencies	US-18.1; Patient Care & Medical Record Management; Secure Document Storage; Accounts queue
Definition of Done	• Side-by-side claim/patient/proof review is implemented. • Accept/Reject actions are explicit; rejection reason is mandatory. • Accepted claim routing to Accounts and blocked rejected-claim payment are tested.

Story ID	AC ID	Acceptance Criteria
US-18.3	AC-88	Given: a validated reimbursement claim awaiting CMO review When: the CMO completes the review Then: Accept records the decision/approved amount and routes the claim to Accounts, while Reject requires a reason and prevents the claim from proceeding to payment.

4.5.2 EPIC-19: Referral Billing Management

Epic Description and Priority

Epic Field	Description
Description	Manages external-hospital bills/claims received through an approved channel and links each provider claim to the original institutional referral. This is a provider-claim workflow and is distinct from the patient's upfront referral counter payment.
Business Value	Ensures external hospital payouts are accurate and tied to valid institutional referrals.
Priority	High
Target Release	Release 1

User Stories

US-19.1: Record and Verify External Hospital Referral Claim	
ID	US-19.1
Epic	EPIC-19: Referral Billing Management
Story	As an authorized Finance or Medical Center staff member, I want to record a referral bill/claim and supporting documents received from an external hospital, so that the claim can be linked to the original referral, verified, and routed for internal review and payment.
Priority	High
Precondition	- The internal staff member is authenticated and authorized to record referral claims. - The external hospital's bill/claim and required supporting documents have been received through an approved external channel. - A valid referral code or other traceable referral reference is available.

Postcondition	- The external claim is stored and linked to the original referral and supporting documents. - Referral validity is checked and the payable amount is calculated according to the applicable tariff or agreement before CMO review in US-19.2.
Story Points	
Dependencies	Emergency, Diagnostic & Referral Services Referral API; External Referral Claim Intake; Secure Document Storage
Definition of Done	- Internal referral-claim entry and document-linking flow is functional. - Verification against Emergency, Diagnostic & Referral Services referral records is tested. - No external-hospital Medical Center login or submission portal is required.

Story ID	AC ID	Acceptance Criteria
US-19.1	AC-89	Given: an external hospital claim received through an approved channel with a traceable referral reference When: authorized internal staff record and verify it Then: the claim is linked to the original referral and supporting documents and the payable amount is calculated according to the applicable tariff or agreement for subsequent review.

US-19.2: Review and Approve Referral Billing	
ID	US-19.2
Epic	EPIC-19: Referral Billing Management
Story	As the CMO, I want to review a verified external-hospital referral claim and approve or reject it, so that only valid claims proceed to payment.
Priority	High
Precondition	- The CMO is authenticated and authorized for this function. - Required dependencies are available: US-19.1, Emergency, Diagnostic & Referral Services Logs. - A recorded and verified referral claim from US-19.1 is awaiting CMO review.
Postcondition	The claim is approved or rejected with the decision and audit trail preserved.
Story Points	
Dependencies	US-19.1, Emergency, Diagnostic & Referral Services Logs

Definition of Done	Audit views for hospital history operational; Flagging logic for duplicates/suspicious entries functional; Accept/Reject decision buttons active.
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Story ID	AC ID	Acceptance Criteria
US-19.2	AC-90	Given: a verified external-hospital referral claim awaiting CMO review When: the CMO completes the review Then: approval records the decision and allows payment processing, while rejection records a reason and prevents the claim from proceeding to payment.

4.5.3 EPIC-20: Accounts & Payment Processing

Epic Description and Priority

Epic Field	Description
Description	Provides one consolidated Finance workspace for four distinct financial purposes: Approved Payouts (money out after medical/claim approval), Referral Counter Payments (money received upfront from a patient), External Hospital Bills/Claims (provider claim intake linked to the original referral), and Refunds (money returned against an original payment). It also executes eligible bank/payment actions and retains immutable financial audit trails.
Business Value	Provides financial accountability and automates the actual disbursement of funds.
Priority	High
Target Release	Release 2

User Stories

US-20.1: Manage Consolidated Finance Workspace	
ID	US-20.1
Epic	EPIC-20: Accounts & Payment Processing
Story	As an Accounts Officer, I want one consolidated Finance workspace that clearly separates approved payouts, upfront referral counter payments, external-hospital bills/claims, and refunds, so that money-in, money-out, provider claims, and reversals are not confused or duplicated.
Priority	High

Precondition	- The Accounts Officer is authenticated and authorized for this function. - Relevant source workflows are available: EPIC-10 referral payment/refund, EPIC-18 reimbursement, and EPIC-19 external-hospital claims.
Postcondition	- The Finance workspace presents Approved Payouts, Referral Counter Payments, External Hospital Bills/Claims, and Refunds as clearly separated sections with their own source references and governed statuses. - Only items that are actually eligible for payout are selectable for payment execution/batching; medical approval is not repeated in Accounts.
Story Points	
Dependencies	EPIC-10, EPIC-18, EPIC-19; Payment / Banking API; Audit Logging Service
Definition of Done	- The four finance sections are visually and functionally distinct and do not duplicate one another. - Governed status updates and eligible payout selection are functional. - Accounts does not repeat CMO medical decisions and no redundant finance workspace is introduced.

Story ID	AC ID	Acceptance Criteria
US-20.1	AC-91 [Derived]	Given: an authenticated Accounts Officer When: the Finance workspace is opened Then: approved payouts, referral counter payments, external-hospital bills/claims, and refunds are presented as distinct workflow sections, and eligible approved payouts can be selected for payment execution.

US-20.2: Execute Bank Transfers	
ID	US-20.2
Epic	EPIC-20: Accounts & Payment Processing
Story	As an Accounts Officer, I want to generate and send bank instructions via API, so that funds are electronically transferred to claimants and statuses are updated to "Paid".
Priority	High
Precondition	- The Accounts Officer is authenticated and authorized for this function. - Required dependencies are available: US-20.1, Bank API.
Postcondition	Bank instruction payload generation verified; API transmission acknowledgment logic operational; automated status update to "Paid" on success.
Story Points	

Dependencies	US-20.1, Bank API
Definition of Done	Bank instruction payload generation verified; API transmission acknowledgment logic operational; automated status update to “Paid” on success.

Story ID	AC ID	Acceptance Criteria
US-20.2	AC-92 [Derived]	Given: an authenticated Accounts Officer and a valid payment batch When: bank transfer instructions are submitted Then: the instruction payload is transmitted through the approved bank interface and successfully acknowledged transfers update the corresponding claims to Paid.

US-20.3: Manage Audit Trails	
ID	US-20.3
Epic	EPIC-20: Accounts & Payment Processing
Story	As an Accounts Officer, I want to view and export immutable audit logs, so that I can ensure compliance with financial governance standards.
Priority	High
Precondition	- The Accounts Officer is authenticated and authorized for this function. - Required dependencies are available: All Transactional Stories.
Postcondition	Read-only log viewer implemented; Export (PDF/CSV) functionality verified.
Story Points	
Dependencies	All Transactional Stories
Definition of Done	Read-only log viewer implemented; Export (PDF/CSV) functionality verified.

Story ID	AC ID	Acceptance Criteria
US-20.3	AC-93 [Derived]	Given: an authenticated Accounts Officer with audit permission When: financial audit trails are opened Then: the logs are displayed read-only and permitted PDF/CSV export is available without altering the source records.

4.5.4 EPIC-21: Procurement Management

Epic Description and Priority

Epic Field	Description
Description	Continues the procurement process after the CMO identifies and submits a replenishment need in US-16.2. All subsequent administrative, VC-approval, tender/vendor and delivery steps are presented within the combined Procurement / Authorized Admin / VC workspace; each user sees only the actions permitted to their authorization. Accepted delivery quantities are handed back to inventory once.
Business Value	Creates a traceable separation of responsibilities from clinical stock need to administrative submission, executive approval, tendering, delivery, and synchronized inventory update.
Priority	High
Target Release	Release 2

User Stories

US-21.1: Submit Procurement Request for VC Review	
ID	US-21.1
Epic	EPIC-21: Procurement Management
Story	As an authorized administration user within the Procurement workspace, I want to review a procurement request received from the CMO and submit it for VC review, so that the request enters the formal approval process with complete supporting information.
Priority	High
Precondition	- The administrator is authenticated and authorized to process procurement requests. - A CMO procurement request from US-16.2 exists with status Pending Administrative Submission.
Postcondition	- A complete request is submitted to the VC and its status becomes Awaiting VC Approval. - The submitting administrator and timestamp are recorded. - An incomplete request remains pending and the missing information is identified.
Story Points	
Dependencies	US-16.2 Submit Procurement Request; Role-Based Access Control; Audit Logging Service
Definition of Done	- Administrative submission and completeness validation are functional. - Handoff to the VC approval queue and audit trace are tested.

Story ID	AC ID	Acceptance Criteria
US-21.1	AC-94 [Derived]	Given: a CMO procurement request that is ready for administrative submission When: an authorized administration user submits it for VC review Then: a complete request moves to Awaiting VC Approval and records the submitting administrator and timestamp; an incomplete request is blocked and the missing information is identified.

US-21.2: Review and Approve Procurement Request	
ID	US-21.2
Epic	EPIC-21: Procurement Management
Story	As a VC-authorized user within the Procurement workspace, I want to approve or reject a procurement request submitted for my review, so that only authorized requests proceed to tender.
Priority	High
Precondition	- The user has VC-level procurement approval permission. - A procurement request from US-21.1 exists with status Awaiting VC Approval.
Postcondition	- An approved request changes to Approved and becomes eligible for tender creation. - A rejected request records a mandatory reason and does not proceed to tender. - The decision and timestamp are retained.
Story Points	
Dependencies	US-21.1 Submit Procurement Request for VC Review; Role-Based Access Control; Audit Logging Service
Definition of Done	- VC approval/rejection flow is functional. - Decision logging and rejection-reason validation are tested.

Story ID	AC ID	Acceptance Criteria
US-21.2	AC-95	Given: a procurement request with status Awaiting VC Approval When: a VC-authorized user reviews the request Then: approval changes the status to Approved, logs the decision, and enables tender creation; rejection requires a reason and does not allow the request to proceed to tender.

US-21.3: Manage Tender Lifecycle	
ID	US-21.3

Epic	EPIC-21: Procurement Management
Story	As a procurement-authorized user within the Procurement workspace, I want to create, track, and finalize a tender for an approved procurement request, so that an authorized vendor can be selected for supply delivery.
Priority	High
Precondition	- The user has procurement/tender permission. - An approved procurement request exists from US-21.2.
Postcondition	- The tender lifecycle is recorded against the approved request. - A finalized tender identifies the selected vendor and preserves the decision history required for delivery processing.
Story Points	
Dependencies	US-21.2 Review and Approve Procurement Request
Definition of Done	- Tender creation, tracking, and vendor finalization are functional. - Linkage to the approved procurement request is verified.

Story ID	AC ID	Acceptance Criteria
US-21.3	AC-96 [Derived]	Given: an approved procurement request When: a procurement-authorized user creates and completes the tender process Then: tender publication and tracking are recorded and the selected vendor is linked to the approved request with traceable decision history.

US-21.4: Record Delivery and Update Stock	
ID	US-21.4
Epic	EPIC-21: Procurement Management
Story	As a delivery-authorized user within the Procurement workspace, I want to record delivery against the finalized procurement and pass the accepted quantities to inventory, so that digital stock records reflect the received supply.
Priority	High
Precondition	- The administrator is authenticated and authorized for delivery recording. - A finalized tender and selected vendor exist from US-21.3. - A physical delivery has been received and its accepted quantities are available for entry.

Postcondition	- The delivery record is linked to the procurement request, tender, vendor, and accepted quantities. - Accepted quantities are sent to Pharmacy, Inventory & Laboratory Management and inventory is updated once without duplicating the delivery transaction.
Story Points	
Dependencies	US-21.3 Manage Tender Lifecycle; Pharmacy, Inventory & Laboratory Management inventory-update interface
Definition of Done	- Delivery linkage and accepted-quantity validation are tested. - Inventory update is integration-tested for success and duplicate/retry protection.

Story ID	AC ID	Acceptance Criteria
US-21.4	AC-97	Given: a received delivery for a finalized procurement When: a delivery-authorized user confirms the accepted quantities Then: the delivery is recorded and the accepted quantities are sent once to the inventory-update interface so stock reflects the received supply.

4.6 Administrative, Reporting & Scheduling System

This subsection covers duty-roster planning, changes/cover, schedule visibility, night on-call operations, formal laboratory service-count reporting and yearly policy-adherence audit. The audit is available only to an explicitly Accounts / Finance Officer with audit permission; no separate dedicated audit-office application role is required.

4.6.1 EPIC-22: Duty Roster Planning and Publication

Epic Description and Priority

Epic Field	Description
Description	The CMO plans duty for a period and publishes it. The same workspace produces the weekly staff roster for configured doctors and nurses/medical attendants and the night on-call roster that identifies who answers the hotline. The system checks every draft for clashes before it can go live, and tells each staff member whose own duty changed.
Business Value	The centre is covered 24 hours a day, nobody is put on two shifts at once, and nobody learns about a duty change by word of mouth. This replaces the paper noticeboard and the verbal relay used today.
Source Use Case(s)	UC-87
Priority	High

Epic Field	Description
Target Release	Release 1

User Stories

US-22.1: Create and Publish the Weekly Duty Roster	
ID	US-22.1
Epic	EPIC-22: Duty Roster Planning and Publication
Story	As the CMO, I want to draft a weekly duty roster and have every clash reported before I publish, so that the centre is staffed around the clock without double-booked or empty shifts.
Priority	High
Precondition	- The CMO is authenticated and authorized to manage duty rosters. - Staff list, staff availability, and shift timings are configured for the target period.
Postcondition	- A clash-free, fully covered roster may be published for the period. - Drafts with uncovered shifts remain drafts with warnings; overlapping assignments or missing availability prevent publication.
Story Points	
Dependencies	EPIC-1 Authentication and Session Management; configured staff list, availability, and shift timings.
Definition of Done	Code reviewed, unit tested, acceptance criteria met, clash rules documented.

Story ID	AC ID	Acceptance Criteria
US-22.1	AC-98	Given: a draft that places one person on two overlapping shifts When: the CMO tries to publish Then: publication is refused and the clash is reported.
US-22.1	AC-99	Given: a draft with a shift nobody is on When: it is saved Then: it can be kept only as a draft carrying a warning, never published.
US-22.1	AC-100	Given: a clash-free draft When: the CMO publishes Then: the schedule goes live for the period.

Story ID	AC ID	Acceptance Criteria
US-22.1	AC-101	Given: missing staff availability data When: the check runs Then: it cannot finish and the CMO is warned.

US-22.2: Create and Publish the Night On-Call Roster	
ID	US-22.2
Epic	EPIC-22: Duty Roster Planning and Publication
Story	As the CMO, I want a night on-call roster naming one doctor and one nurse for every night, so that someone is always reachable when no doctor is on site.
Priority	High
Precondition	- The CMO is authenticated and authorized to manage the on-call roster. - The day roster for the target period exists.
Postcondition	- Every published night has the required on-call coverage without a clash with the same person's day shift. - The published on-call list feeds the public on-duty board at night.
Story Points	
Dependencies	US-22.1; the published day roster for the period.
Definition of Done	Code reviewed, unit tested, acceptance criteria met, on-call eligibility rules documented.

Story ID	AC ID	Acceptance Criteria
US-22.2	AC-102	Given: a period in which any night has no on-call doctor When: publication is attempted Then: it is refused, the gap is logged and it is sent to the CMO.
US-22.2	AC-103	Given: an on-call assignment that clashes with the same person's day shift When: the draft is checked Then: the assignment is rejected.
US-22.2	AC-104	Given: a published on-call roster When: night falls Then: the on-call list feeds the public on-duty board.

US-22.3: Notify Staff Whose Own Duty Changed
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ID	US-22.3
Epic	EPIC-22: Duty Roster Planning and Publication
Story	As a staff member, I want a message on my preferred channel whenever my own duty changes, so that I never find out about a change by word of mouth.
Priority	High
Precondition	- A new roster has been published and can be compared with the previous published roster. - Staff contact details and preferred notification channels are available.
Postcondition	- Each staff member whose own duty changed receives exactly one notification and the delivery result is recorded. - No duty-change message is sent when a staff member's own duty did not change; failed delivery is retried or routed to an approved alternate channel.
Story Points	
Dependencies	US-22.1; staff contact details and preferred notification channel on record.
Definition of Done	Code reviewed, unit tested, acceptance criteria met, channel preference handling documented.

Story ID	AC ID	Acceptance Criteria
US-22.3	AC-105	Given: a newly published roster When: the system compares it with the previous one Then: every staff member whose own duty changed receives exactly one message and the delivery result is recorded.
US-22.3	AC-106	Given: a publication in which nobody's own shift changed When: it goes live Then: no message is sent.
US-22.3	AC-107	Given: a message that fails to deliver When: the failure is detected Then: it is retried or sent on an alternate channel.

4.6.2 EPIC-23: Duty Changes, Absence and Cover

Epic Description and Priority

Epic Field	Description
Description	A duty change travels from the moment it is asked for to the moment it is decided, entirely on record. A staff member requests a swap, leave or absence; the CMO decides it; a replacement of the same role is placed before the change takes effect. The Registrar records an absence or late arrival, and the CMO uses the same-role replacement steps to restore coverage.
Business Value	No shift is given up until someone else holds it, and it is the shift that gets covered, not the person first put on it. Refusals always carry a reason, so staff know why a change was not allowed.
Source Use Case(s)	UC-89, UC-90
Priority	High
Target Release	Release 1

User Stories

US-23.1: Request a Shift Change	
ID	US-23.1
Epic	EPIC-23: Duty Changes, Absence and Cover
Story	As a staff member, I want to ask to be taken off a future shift with a stated reason, so that a change I need is decided on record rather than settled verbally.
Priority	High
Precondition	- The staff member is authenticated and the selected shift is a future shift on the current published roster. - No other open request exists for the same shift.
Postcondition	- A valid request containing type and reason is placed in the CMO's decision list and the CMO is alerted. - A withdrawn undecided request is closed without a decision; requests for started or past shifts are refused.
Story Points	
Dependencies	EPIC-22; the shift is on the current published roster.
Definition of Done	Code reviewed, unit tested, acceptance criteria met, request states documented.

Story ID	AC ID	Acceptance Criteria
US-23.1	AC-108	Given: a future rostered shift When: the staff member submits a request with a type (swap, leave or absence) and a reason Then: the request lands in the CMO's list and he is alerted.
US-23.1	AC-109	Given: a shift that has already started or passed When: a request on it is submitted Then: it is refused.
US-23.1	AC-110	Given: an open request on a shift When: a second request on the same shift arrives Then: it is blocked.
US-23.1	AC-111	Given: an undecided request When: the requester withdraws it Then: it is closed without a decision.

US-23.2: Decide a Shift Change Request	
ID	US-23.2
Epic	EPIC-23: Duty Changes, Absence and Cover
Story	As the CMO, I want to approve or reject each request and place a replacement in the same step, so that no shift is given up before someone else holds it.
Priority	High
Precondition	- The CMO is authenticated and an undecided shift-change request exists. - The published roster and free-staff list for the same role are available.
Postcondition	- An approval assigns a same-role replacement, preserves the previous assignment in history, and records who decided and when. - A rejection requires a reason and leaves the roster unchanged; if nobody is free, the shift remains marked unfilled on the CMO's list.
Story Points	
Dependencies	US-23.1.
Definition of Done	Code reviewed, unit tested, acceptance criteria met, decision history documented.

Story ID	AC ID	Acceptance Criteria
US-23.2	AC-112	Given: an approval When: the decision is saved Then: a replacement of the same role from the free list holds the shift, the previous name stays in history, and the decision is stored with who made it and when.
US-23.2	AC-113	Given: a rejection with no reason typed When: the CMO tries to save it Then: the rejection cannot be saved.
US-23.2	AC-114	Given: a rejection with a reason When: it is saved Then: the roster is left unchanged and the requester is told.
US-23.2	AC-115	Given: nobody free to replace When: the approval is attempted Then: the shift is marked unfilled and remains on the CMO's list.

US-23.3: Mark a Staff Member Absent or Late and Restore Cover	
ID	US-23.3
Epic	EPIC-23: Duty Changes, Absence and Cover
Story	As a Registrar Office user, I want to record an absence or late arrival and hand the coverage requirement to the CMO, so that a suitable same-role replacement can be assigned and the gap is visible immediately.
Priority	High
Precondition	- The Registrar Office user is authenticated and the affected shift has not finished. - The rostered staff member and the applicable absence or lateness period are identified.
Postcondition	- The outcome is either a same-role replacement or a gap reported to the CMO. - Late-arrival cover ends when the rostered person reports; a multi-day absence is stored as one period entry.
Story Points	
Dependencies	EPIC-22; reuses the same-role replacement steps from US-23.2.
Definition of Done	Code reviewed, unit tested, acceptance criteria met, free-staff rules documented.

Story ID	AC ID	Acceptance Criteria
US-23.3	AC-116	Given: a rostered staff member marked absent When: the record saves Then: the outcome is either a named replacement of the same role or a gap reported to the CMO.
US-23.3	AC-117	Given: a late arrival When: a replacement is placed Then: the cover ends when the rostered person reports for duty.
US-23.3	AC-118	Given: a shift that has already finished When: the Registrar tries to mark it Then: the action is blocked.
US-23.3	AC-119	Given: an absence spanning several days When: it is recorded Then: one entry covers the whole period.

4.6.3 EPIC-24: Schedule Visibility

Epic Description and Priority

Epic Field	Description
Description	Two read-only views of the published rosters. A staff member sees their own shifts and on-call nights without walking to the office. Students and anyone in the waiting area see who is on duty right now and the emergency hotline, with no sign-in needed. The board shows no patient detail and no clinical detail.
Business Value	The paper noticeboard goes away. Staff and students read the schedule wherever they are, and the board is the one screen deliberately open to everyone.
Source Use Case(s)	UC-88, UC-91
Priority	High
Target Release	Release 1

User Stories

US-24.1: View My Own Duty Schedule	
ID	US-24.1
Epic	EPIC-24: Schedule Visibility

Story	As a staff member, I want to see my upcoming shifts and on-call nights and anything that changed since I last looked, so that I do not have to read a noticeboard at the office.
Priority	Medium
Precondition	- The staff member is authenticated. - A published roster may exist for the selected period.
Postcondition	- The user sees only their own published shifts, on-call nights, and recent changes in read-only form. - A listed future shift can open the shift-change request flow; when no roster exists, the view shows an empty state and the last published date.
Story Points	
Dependencies	EPIC-22.
Definition of Done	Code reviewed, unit tested, acceptance criteria met.

Story ID	AC ID	Acceptance Criteria
US-24.1	AC-120	Given: a published roster When: a staff member opens their schedule Then: their own shifts and recent changes are shown read-only, and nobody else's duty is visible.
US-24.1	AC-121	Given: nothing published for the period When: the schedule is opened Then: an empty view is shown carrying the last published date.
US-24.1	AC-122	Given: a listed shift When: the staff member selects it Then: a shift change request can be raised directly from it.

US-24.2: View the Public On-Duty Board	
ID	US-24.2
Epic	EPIC-24: Schedule Visibility
Story	As a student, I want to see who is on duty right now and the emergency hotline without signing in, so that I know before walking to the centre.
Priority	High

Precondition	- The public board is available without authentication and the hotline number is configured. - Published duty/on-call data may exist for the current time.
Postcondition	- The board shows the current doctor and nurse, shift end and hotline when applicable; at night it identifies the on-call doctor and states that care is provided over the phone. - If no roster covers the current time, only the hotline and an on-call note are shown; roster changes appear on refresh.
Story Points	
Dependencies	EPIC-22; approved hotline number configured.
Definition of Done	Code reviewed, unit tested, acceptance criteria met, waiting-area screen refresh documented.

Story ID	AC ID	Acceptance Criteria
US-24.2	AC-123	Given: the current time falls inside a published shift When: the board is opened Then: the doctor and nurse on duty, the shift end and the hotline are shown.
US-24.2	AC-124	Given: night hours When: the board is opened Then: the on-call doctor is named and the board states that care is given over the phone.
US-24.2	AC-125	Given: nothing published for the current time When: the board is opened Then: only the hotline is shown with an on-call note.
US-24.2	AC-126	Given: a roster change When: the next refresh runs Then: the board reflects it.

4.6.4 EPIC-25: Night On-Call Operations

Epic Description and Priority

Epic Field	Description
Description	No doctor is on site at night. The on-call doctor gives first-aid advice over the phone, and every call is recorded against the on-call shift. When advice is not enough, the on-call workflow creates a transfer handoff linked to the call and reuses the existing Emergency ambulance request and live-tracking workflow (US-7.2/US-7.3). The destination and clinical reason are carried forward, an authorized on-duty doctor or nurse issues the request directly to an eligible ambulance driver, the driver accepts or rejects it, and accepted trips expose live location/status to the requesting clinical staff. Administrative, Reporting & Scheduling does not create a second ambulance process.
Business Value	Night care leaves a record, and a student who cannot be helped on the phone is moved to hospital without delay. No call is lost, and a request that finds no driver always produces an alert and a logged gap.
Source Use Case(s)	UC-92
Priority	High
Target Release	Release 1

User Stories

US-25.1: Log a Night On-Call Call	
ID	US-25.1
Epic	EPIC-25: Night On-Call Operations
Story	As the on-call doctor, I want each hotline call logged with the caller, the problem, the advice and the outcome, so that night care at the centre leaves a record.
Priority	High
Precondition	- A published night on-call roster identifies the on-call doctor for the current period. - A hotline call has been received.
Postcondition	- One call log is linked to the correct on-call shift with caller identifier when known, on-call staff, time, problem, advice and outcome. - Unknown callers may be recorded by phone number only; failed saves are retained for retry and morning follow-up flags reach the day shift.
Story Points	
Dependencies	US-22.2; a published on-call roster.
Definition of Done	Code reviewed, unit tested, acceptance criteria met, outcome codes documented.

Story ID	AC ID	Acceptance Criteria
US-25.1	AC-127	Given: a night call When: the doctor records it Then: one log entry exists on the correct on-call shift with the caller's IUT ID, the on-call staff, the time, the problem, the advice and the outcome.
US-25.1	AC-128	Given: a caller who cannot be identified When: the call is logged Then: it is recorded with the phone number only.
US-25.1	AC-129	Given: a log entry that fails to save When: the failure occurs Then: the entry is kept and retried, and no call is lost.
US-25.1	AC-130	Given: a case flagged for the morning When: the day shift starts Then: the flag reaches the day shift.

US-25.2: Escalate a Night Case to an Ambulance Transfer	
ID	US-25.2
Epic	EPIC-25: Night On-Call Operations
Story	As the on-call doctor, I want to hand a logged night case that requires transfer to the existing ambulance request and live-tracking workflow, so that the same direct doctor/nurse-to-driver process is reused and no duplicate ambulance workflow is created.
Priority	High
Precondition	- A night call has been logged and the on-call doctor determines that transfer is required. - A destination, priority, clinical reason and handover information are available for the ambulance request.
Postcondition	- A transfer handoff linked to the night-call record is created and passed to US-7.2/US-7.3 so an authorized on-duty doctor or nurse can issue the ambulance request directly to an eligible driver. - Driver acceptance/rejection, live trip status when accepted, no-driver/vehicle exceptions, or self-arranged transport status are written back to the night-call record without creating a second ambulance request/transfer record.
Story Points	
Dependencies	US-25.1; US-7.2 Call Ambulance from Emergency Care; US-7.3 Driver Accepts Request and Shares Live Location; Transport Office driver-duty schedule consumed read-only to identify an eligible driver.
Definition of Done	Code reviewed, unit tested, acceptance criteria met, transfer-handoff and outcome-linkage format documented.

Story ID	AC ID	Acceptance Criteria
US-25.2	AC-131	Given: a logged night case that requires transfer When: the on-call doctor escalates it Then: a transfer handoff linked to the call is created and the existing US-7.2/US-7.3 ambulance request/live-tracking workflow is reused.
US-25.2	AC-132	Given: the Emergency ambulance request workflow reports no eligible driver or vehicle available When: the exception is returned Then: the outcome is written back to the night-call record and the doctor receives the status needed to arrange an alternative.
US-25.2	AC-133	Given: a student who arranges their own transport When: the case closes Then: the call record states “transfer advised – self-arranged” and no duplicate ambulance request/transfer record is created.

4.6.5 EPIC-26: Laboratory Service-Count Reporting

Epic Description and Priority

Epic Field	Description
Description	Produces an administrative service-count report for formal laboratory blood and urine tests processed through EPIC-17, grouped by test type and patient type. It does not count EPIC-12 diagnostic-service bookings, which are covered by the clinical diagnostic-activity summary in EPIC-13. Only authorized count data are used; individual laboratory results are not copied into the report.
Business Value	The centre can state how many formal laboratory blood/urine tests were processed in any period without counting register lines by hand and without holding a second copy of laboratory-result data.
Source Use Case(s)	UC-94 (Generate Diagnostic Test Report)
Priority	Medium
Target Release	Release 2

User Stories

US-26.1: Generate Administrative Laboratory Service-Count Report	
ID	US-26.1

Epic	EPIC-26: Laboratory Service-Count Reporting
Story	As a nurse or medical attendant with reporting permission, I want a count of the formal laboratory blood and urine tests processed over a chosen period, broken down by test type and patient type, so that manual tallying stops.
Priority	Medium
Precondition	- The nurse or medical attendant is authenticated and has reporting permission. - The requested period is valid; diagnostic service records may or may not exist for the period.
Postcondition	- A report is generated from EPIC-17 laboratory service records with counts by blood/urine test type and student/faculty/staff patient type; a period with no tests produces zero counts. - EPIC-12 diagnostic-service activity is excluded from this report; only counts are shown and individual laboratory results are not copied into the report.
Story Points	
Dependencies	EPIC-1 Authentication and Session Management; EPIC-17 formal laboratory service-count records from Pharmacy, Inventory & Laboratory Management; no EPIC-12 activity and no individual result payload.
Definition of Done	Code reviewed, unit tested, acceptance criteria met, report layout agreed with the CMO.

Story ID	AC ID	Acceptance Criteria
US-26.1	AC-134	Given: a valid period When: the report is generated Then: counts from EPIC-17 formal laboratory records appear by test type (blood, urine) and by patient type (student, faculty and staff).
US-26.1	AC-135	Given: a period with no tests When: the report runs Then: a zero-count report is produced, not an error.
US-26.1	AC-136	Given: any generated report When: it is read Then: only EPIC-17 laboratory service counts are shown; EPIC-12 diagnostic-service activity and individual test results are excluded.
US-26.1	AC-137	Given: a filter for one test type When: it is applied Then: the report covers blood only or urine only.

US-26.2: Validate, Store and Retrieve Administrative Diagnostic Reports	
ID	US-26.2
Epic	EPIC-26: Laboratory Service-Count Reporting
Story	As the Chief Medical Officer, I want every generated report validated, stored and retrievable, so that any past report can be traced to who produced it and when.
Priority	Medium
Precondition	A permitted laboratory service-count report request has been generated for a non-future period.
Postcondition	- The report is stored with period, generating user and generation time and remains retrievable within role limits. - Regeneration creates a timestamped new version while retaining the earlier version for traceability; unauthorized report access is refused and recorded.
Story Points	
Dependencies	US-26.1.
Definition of Done	Code reviewed, unit tested, acceptance criteria met, retention documented.

Story ID	AC ID	Acceptance Criteria
US-26.2	AC-138	Given: a future date When: a report on it is requested Then: the request is rejected.
US-26.2	AC-139	Given: a generated report When: it is stored Then: it carries its period, its generating user and its generation time, and remains retrievable.
US-26.2	AC-140	Given: a report regenerated for a past period When: the new version is saved Then: it is stored as a new timestamped version while the earlier version remains retained and retrievable for traceability.
US-26.2	AC-141	Given: a requesting role without reporting permission When: a report is requested Then: access is refused and the refusal is recorded.

4.6.6 EPIC-27: Policy Adherence Audit

Epic Description and Priority

Epic Field	Description
Description	Once a year an Accounts / Finance Officer with audit permission audits the year's reimbursement records against the approved reimbursement policy. The system retrieves every record for the selected year, compares each against the configured policy rules, and highlights the ones that exceed or violate a limit. The audit checks adherence; it does not total yearly expenditure. Claim processing remains within Financial, Billing & Reimbursement Management, whose authorized records are read for the audit.
Business Value	Violations surface for review instead of staying buried in the records, and each yearly audit leaves a stored result and report.
Source Use Case(s)	UC-98
Priority	High
Target Release	Release 3

User Stories

US-27.1: Run the Yearly Policy Adherence Audit	
ID	US-27.1
Epic	EPIC-27: Policy Adherence Audit
Story	As an Accounts / Finance Officer with audit permission, I want the year's reimbursement records checked against the approved reimbursement policy, so that violations surface for review instead of staying buried in the records.
Priority	High
Precondition	- The Accounts / Finance Officer is authenticated and has explicit audit permission. - The selected year, reimbursement records and configured policy rules are available.
Postcondition	- Every reimbursement record for the selected year is compared with the configured policy rules and violations are highlighted. - If required records or policy configuration are missing, the audit aborts and reports what is missing; output reports adherence only, not yearly expenditure totals.
Story Points	
Dependencies	Financial, Billing & Reimbursement Management reimbursement records (read-only); configured reimbursement policy rules.
Definition of Done	Code reviewed, unit tested, acceptance criteria met, policy rule configuration documented.

Story ID	AC ID	Acceptance Criteria
US-27.1	AC-142	Given: a selected year When: the audit runs Then: every reimbursement record for that year is compared against the policy rules and each violation is highlighted.
US-27.1	AC-143	Given: missing reimbursement records or missing policy configuration When: the audit is started Then: it aborts and reports what is missing.
US-27.1	AC-144	Given: the audit output When: it is read Then: it shows policy adherence only and no yearly expenditure total.

US-27.2: Store the Audit Result and Report	
ID	US-27.2
Epic	EPIC-27: Policy Adherence Audit
Story	As an Accounts / Finance Officer with audit permission, I want the audit result stored and an audit report generated, so that the finding of each yearly audit stays on record.
Priority	High
Precondition	A yearly policy-adherence audit has completed successfully.
Postcondition	- The audit result is stored and an audit report is generated and retained. - A year with no violations is stored as a clean audit result rather than treated as an error.
Story Points	
Dependencies	US-27.1.
Definition of Done	Code reviewed, unit tested, acceptance criteria met, audit report format documented.

Story ID	AC ID	Acceptance Criteria
US-27.2	AC-145	Given: a completed audit When: it finishes Then: the result is stored and an audit report is generated.
US-27.2	AC-146	Given: a year with no violations When: the audit completes Then: a clean result is recorded, not an error.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

ID	Requirement	Metric / Target	Mapped User Stories
NFR-01	Validate login and display the next permitted page.	≤ 1 second under normal conditions.	US-1.1
NFR-02	Exact patient-ID search.	≤ 1 second.	US-2.1, US-3.1, US-6.1–US-6.3
NFR-03	Patient-name search first results page.	≤ 2 seconds.	US-2.1, US-3.1, US-6.1–US-13.1
NFR-04	Validate and save a patient-care record.	≤ 1.5 seconds.	US-2.1
NFR-05	Save a valid prescription draft.	≤ 1.5 seconds.	US-3.1
NFR-06	Finalize a prescription and show Active status.	≤ 2 seconds.	US-3.2
NFR-07	Display active prescription.	≤ 1.5 seconds.	US-3.3, US-14.2–US-14.3
NFR-08	Display first page of prescription history.	≤ 2 seconds.	US-3.4
NFR-09	Save a valid daily-tracking entry.	≤ 1.5 seconds.	US-2.2
NFR-10	Save a lab request and place it in the pending queue.	≤ 2 seconds.	US-4.1, US-17.1
NFR-11	Open an authorized operational record/report list and submit a selected item to the CMO routing queue.	≤ 3 seconds under normal conditions.	US-2.4
NFR-12	Generate monthly medicine-bill report for up to 1,000 transactions.	≤ 8 seconds.	US-2.5
NFR-13	Generate or download an authorized PDF/Excel/report document up to 50 pages.	≤ 10 seconds.	US-2.4, US-2.5, US-9.3, US-13.1, US-16.1, US-17.3, US-20.3, US-26.1, US-27.2
NFR-14	Validate and store PDF/JPG/JPEG/PNG up to 10 MB over ≥ 10 Mbps.	≤ 10 seconds.	US-2.3, US-9.2, US-18.1, US-19.1
NFR-15	Create internal notification after key account/report/certificate/exam events.	≤ 5 seconds.	US-5.1–US-5.3, US-9.4, US-22.3, US-23.1–US-23.3

ID	Requirement	Metric / Target	Mapped User Stories
NFR-16	Support daily emergency/referral/diagnostic volume.	50–70 registrations per day.	US-6.1–US-13.1
NFR-17	Support simultaneous peak-hour use.	Configured peak-hour clinical, laboratory and Accounts/Finance users concurrently, including at least 3 doctors and 2 nurse/medical-attendant sessions.	US-2.1, US-4.1, US-6.1–US-13.1, US-17.1–US-20.3
NFR-18	Operational availability.	24/7; target $\geq 99\%$ uptime.	US-6.1–US-8.1, US-22.1–US-25.2
NFR-19	Authorized patient-record retrieval.	≤ 3 seconds under normal load.	US-2.1–US-5.3, US-6.1–US-17.3

5.2 Safety Requirements

ID	Safety Requirement	Safeguard	Mapped User Stories
NFR-20	Emergency care shall never be delayed for administrative completeness.	Fallback/provisional registration and continued triage when identity is unresolved.	US-6.1–US-7.1
NFR-21	Implausible vital-sign entries shall not silently pass.	Range validation and visible staff warning while preserving emergency workflow.	US-2.1, US-7.1
NFR-22	Bed-count mismatch shall not cause unsafe over-allocation.	Logged manual discrepancy correction and confirmation before reservation.	US-8.1
NFR-23	Expired, unavailable, or insufficient medicine shall not be dispensed as fully available.	Availability check, batch/expiry control, partial-dispensing record, and stock count/adjustment control.	US-14.1–US-15.3
NFR-24	Unverified laboratory results shall not be published as final.	Formal verification gate, status control, and audit log.	US-17.2–US-17.3
NFR-25	Financial execution shall remain separate from medical validity approval.	Role separation between CMO/clinical reviewer and Accounts/Finance execution.	US-10.1–US-10.3, US-18.1–US-20.2

5.3 Security Requirements

ID	Requirement	Mapped User Stories
NFR-26	Access shall be granted only when an official IUT email and password match an active authorized account.	US-1.1; all protected user stories
NFR-27	An account marked First Time Login shall be restricted to password change until successfully completed.	US-1.2–US-1.3
NFR-28	An account shall be temporarily blocked for 15 minutes after 5 consecutive failed login attempts within 15 minutes.	US-1.1
NFR-29	Role-based access control and least privilege shall apply to every workspace and sensitive action.	All protected user stories
NFR-30	Emergency, clinical, referral, diagnostic, inventory, laboratory, financial, procurement, and audit events shall record responsible user and timestamp.	Audit-relevant stories across US-2.1–US-27.2
NFR-31	All data in transit shall use TLS/HTTPS or another ICT-approved secure protocol.	All web/service-integrated user stories
NFR-32	Patient and financial data shall not be exposed in general reports, logs, or notification bodies.	US-2.4–US-2.5, US-13.1, US-20.3, US-24.2, US-26.1–US-27.2
NFR-33	Uploaded documents shall be type/size/readability validated and stored with controlled access and version preservation.	US-2.3, US-5.1–US-5.3, US-9.2–US-10.3, US-17.2–US-19.2
NFR-34	Session expiry, logout, token invalidation, and access revocation shall be enforced by shared security services.	US-1.1–US-1.4; all authenticated user stories
NFR-35	Financial audit logs shall be immutable/read-only to normal operational users.	US-20.3, US-27.1–US-27.2

5.4 Software Quality Attributes

Attribute	Requirement / Target
Usability	Emergency screens shall support one-handed use, large touch targets, minimal required fields, plain-language errors, clear status wording, and a readable map-based live ambulance view after driver acceptance.
Reliability	Interrupted emergency entries should be recoverable; critical operational data shall be backed up daily and reconciled after interface failures.
Interoperability	All cross-subsystem calls shall use documented contracts and be integration/contract tested.

Attribute	Requirement / Target
Maintainability	Clinical, emergency, pharmacy, laboratory, finance, procurement, scheduling, and reporting concerns shall remain modular.
Testability	Each story shall have testable postconditions, acceptance criteria, and a Definition of Done.
Correctness	Final records shall be amended through controlled corrections rather than destructive edits.
Accessibility	Responsive interfaces shall use readable typography, keyboard-operable controls where practical, sufficient contrast, and meaningful validation.
Auditability	Material decisions, status changes, approvals, payments, and exceptions shall be traceable to actor and time.

5.5 Business Rules

BR ID	Rule	Affected Scope
BR-01	Emergency care is free of cost. Referral and diagnostic services require upfront payment under the supplied “Pay First, Get Refunded Later” policy unless an approved exception applies.	Emergency, Diagnostic & Referral Services; Financial, Billing & Reimbursement Management
BR-02	For a standard ambulance transfer, the on-duty doctor or nurse selects/records one of the four designated partner hospitals when calling the ambulance. Another destination requires management approval and a logged exception. The driver accepts/rejects the request but does not make the clinical transfer decision.	Emergency, Diagnostic & Referral Services; Administrative, Reporting & Scheduling System
BR-03	The on-site diagnostic services defined in Emergency, Diagnostic & Referral Services are performed only during the 08:00–14:00 operating window.	Emergency, Diagnostic & Referral Services
BR-04	A referral letter requires a documented doctor recommendation, and only an authorized doctor may initiate the referral.	Patient Care & Medical Record Management; Emergency, Diagnostic & Referral Services
BR-05	An unmatched identity is marked Unverified – Guest or Pending Verification rather than denied emergency care.	Patient Care & Medical Record Management; Emergency, Diagnostic & Referral Services

BR ID	Rule	Affected Scope
BR-06	Final prescriptions, verified lab results, financial transactions, and approved claims cannot be silently overwritten.	Patient Care & Medical Record Management; Pharmacy, Inventory & Laboratory Management; Financial, Billing & Reimbursement Management
BR-07	Medicine dispensing shall be tied to a valid prescription or approved emergency-care record and shall update inventory.	Patient Care & Medical Record Management; Pharmacy, Inventory & Laboratory Management
BR-08	Clinical validity approval and financial payment execution shall be performed by separated authorized roles.	Financial, Billing & Reimbursement Management
BR-09	External hospitals, specialists, and labs do not receive general Medical Center system access. They receive only the information required for an approved transaction/referral, and external claims or documents requiring system entry are recorded by authorized internal IUT staff.	Emergency, Pharmacy/Laboratory, Finance, and Administration subdomains
BR-10	A weekly duty roster shall not be published while it contains overlapping assignments, an uncovered shift, or missing staff-availability data required for validation.	Administrative, Reporting & Scheduling System
BR-11	Every published night on-call roster shall provide the required doctor and nurse coverage, and an on-call assignment shall not conflict with the same person's day shift.	Administrative, Reporting & Scheduling System
BR-12	An approved shift change or absence-cover action shall use a same-role replacement where available; rejection requires a reason, and an unfilled shift shall remain visible for CMO review.	Administrative, Reporting & Scheduling System
BR-13	The public on-duty board shall show duty/on-call and hotline information only; it shall not expose patient or clinical detail.	Administrative, Reporting & Scheduling System
BR-14	EPIC-26 shall count only formal laboratory blood/urine services processed through EPIC-17. EPIC-12 diagnostic-service activity remains in EPIC-13 reporting, and individual result values shall not be duplicated into the administrative count dataset.	Pharmacy, Inventory & Laboratory Management; Emergency, Diagnostic & Referral Services; Administrative, Reporting & Scheduling System
BR-15	The yearly reimbursement audit shall assess adherence to configured policy rules and highlight violations; it shall not substitute for claim processing or present a yearly expenditure total.	Financial, Billing & Reimbursement Management; Administrative, Reporting & Scheduling System

BR ID	Rule	Affected Scope
BR-16	A service follows EPIC-11–EPIC-13 when it is initiated through the diagnostic-service availability/booking/payment workflow. A doctor-issued formal laboratory request follows EPIC-4 and EPIC-17. Routing is determined by the initiating workflow, not by the test name alone.	Patient Care & Medical Record Management; Emergency, Diagnostic & Referral Services; Pharmacy, Inventory & Laboratory Management
BR-17	The Bed Rest / Observation workspace contains exactly six beds in this system scope, and bed status is maintained separately from emergency triage and procedure records.	Emergency, Diagnostic & Referral Services
BR-18	An ambulance request is issued by an authorized on-duty doctor or nurse directly to the eligible driver. After the driver accepts, live location/status is visible to the requesting doctor/nurse until completion; the request goes directly from the clinical caller to the eligible driver.	Emergency, Diagnostic & Referral Services

6. Other Requirements

6.1 Database and Data Requirements

- The patient master record is authoritative in the patient/identity boundary; other subsystems reference it rather than create disconnected duplicate identities.
- Emergency cases, triage, ambulance requests, driver acceptance/location/status events, Bed Rest / Observation assignments, referrals, diagnostic-service order/result, laboratory request/result/report documents, prescriptions, dispensing, inventory, claims, payments, procurement, schedules, notifications, routing events, and audit entities shall use unique identifiers and referential integrity.
- No referral, diagnostic order, emergency case, claim, or financial transaction shall be saved without a valid institutional identifier or an approved temporary emergency identifier.
- Audit-relevant records shall be retained according to IUT policy; emergency, clinical, referral, diagnostic, inventory, and financial data shall be backed up regularly.
- Status vocabularies and master data shall be centrally governed to prevent inconsistent wording across modules.
- Duty rosters, on-call rosters, shift-change requests and decisions, absence/lateness and cover records, night-call logs and ambulance outcomes, generated laboratory service-count reports, CMO record/report routing events, and yearly policy-audit results shall retain the actor, relevant period/time, status, and history needed by their acceptance criteria.

6.2 Internationalization and Presentation

- The primary interface language is English. Bengali labels may be included on patient-facing receipts, referral letters, and notices where practical and approved.
- Amounts shall be displayed in BDT. Dates shall use an unambiguous local format and stored timestamps shall preserve timezone context.
- Printable institutional documents shall use approved headings, references, signatures, and authorization fields.
- Printed or exported prescriptions shall follow the pre-existing approved prescription format(s) as the presentation baseline unless a revised institutional format is formally approved.

6.3 Legal and Policy Requirements

- Referral and hospitalization workflows shall enforce the supplied requirement for prior doctor advice/permission.
- Patient medical, diagnostic, referral, dependent, and financial information shall be treated as confidential institutional data.
- Banking and payment processing shall use approved institutional interfaces and governance controls.

- Procurement approval, tender, vendor selection, and delivery records shall preserve administrative accountability.
- The yearly policy-adherence audit shall use the approved reimbursement policy and authorized reimbursement records, preserve the audit result, and restrict access to permitted roles.

6.4 Reuse Objectives

- Reuse the shared identity, authentication, session, patient record, document storage, notification, reporting, export, and audit services.
- Reuse Financial, Billing & Reimbursement Management payment, receipt, refund, and bank-transfer capabilities rather than duplicating financial logic.
- Reuse Pharmacy, Inventory & Laboratory Management inventory and laboratory functions for availability, deductions, requests, and final results.
- Reuse Administrative, Reporting & Scheduling System duty-roster, shift-change/cover, notification, administrative service-count reporting, and policy-audit capabilities for cross-module coordination.
- Reuse the Emergency ambulance-request/live-tracking workflow for all transfers, including night on-call handoffs. The on-duty doctor/nurse issues the request directly to the eligible driver; the Transport Office duty schedule is read-only, and driver acceptance/location/status updates return to the same source case rather than creating a duplicate ambulance process.

Appendix A: Glossary

Term	Definition
Case ID	Unique identifier assigned to an emergency encounter.
Triage	Clinical severity assessment such as Critical, Urgent, or Non-urgent.
CMO	Chief Medical Officer.
Pay-First Rule	Upfront payment policy for referral and diagnostic services; emergency care is exempt.
Designated Hospital	One of the four partner hospitals approved for standard ambulance transfer.
Ambulance Request	Transfer request issued by an authorized on-duty doctor or nurse directly to the eligible ambulance driver and linked to the source clinical Case ID.
Live Ambulance Tracking	Map-based display of the accepted driver's latest location, last update time, ETA when available, and trip status to authorized requesting clinical staff.
Bed Rest / Observation	Separate six-bed clinical stabilization workspace with Free, Reserved, Occupied, or Cleaning bed states, independent of emergency triage/procedure data entry.
Unverified – Guest	Identity status used when no matching university record is found but emergency registration continues.
Pending Verification	Provisional emergency record awaiting later identity confirmation.
GWT	Given/When/Then acceptance-criteria format.
Priority Scale	High, Medium, and Low prioritization scale used throughout this SRS.
DoD	Definition of Done.
RBAC	Role-Based Access Control.
EFT	Electronic Funds Transfer.
Unified Patient Record	Shared authoritative patient record used across subsystems.

Term	Definition
ICT Account Provisioning	External IUT ICT Center process that creates/maintains the institutional account and authoritative account status consumed by the Medical Center system.
Source Gap	A requirement or story referenced by a supplied source but missing detailed text in the uploaded material.
Duty Roster	Published assignment of Medical Center staff to defined duty shifts for a period.
Night On-Call Roster	Published nightly assignment identifying the doctor and nurse responsible for after-hours hotline coverage.
Public On-Duty Board	Unauthenticated read-only display of current duty/on-call staff, shift timing where applicable, and emergency hotline information without patient or clinical detail.
Shift Change Request	Recorded request to change a future rostered shift through swap, leave, or absence handling and CMO decision.
Policy Adherence Audit	Yearly comparison of authorized reimbursement records against configured reimbursement-policy rules to identify compliance violations.
Designated Higher Financial Authority	University authority that provides additional approval for disputed or over-limit refunds; the exact responsible unit/role is TBD-09.

A.1 Integration and Workflow Model

The system begins with institutional identity/account provisioning by the IUT ICT Center and then moves into Medical Center authentication; the production dashboard is loaded automatically from the authenticated user's role. Routine care uses the unified patient record; finalized prescriptions flow to Pharmacy for dispensing; formal laboratory requests flow to Laboratory operations and verified results return as final viewable/downloadable report documents. Emergency, Diagnostic & Referral Services owns emergency triage, six-bed Bed Rest / Observation, ambulance/referral activity, and its separate diagnostic-service workflow. For ambulance transfer, the on-duty doctor or nurse issues the request directly to the eligible driver; after acceptance, the driver publishes live location/status to the requesting clinical view. Inventory shortages create procurement requests in Pharmacy, while Financial, Billing & Reimbursement Management owns payment execution and procurement workflow. Administrative, Reporting & Scheduling System owns staff rosters, night-call logging, aggregate formal laboratory service-count reporting, and authorized yearly reimbursement-policy auditing. A night transfer does not create a second ambulance process: US-25.2 hands the case to the existing US-7.2/US-7.3 request/live-tracking workflow. External hospitals do not log in; authorized internal staff record claims and supporting documents received through approved channels.

Detailed Workflow Diagram

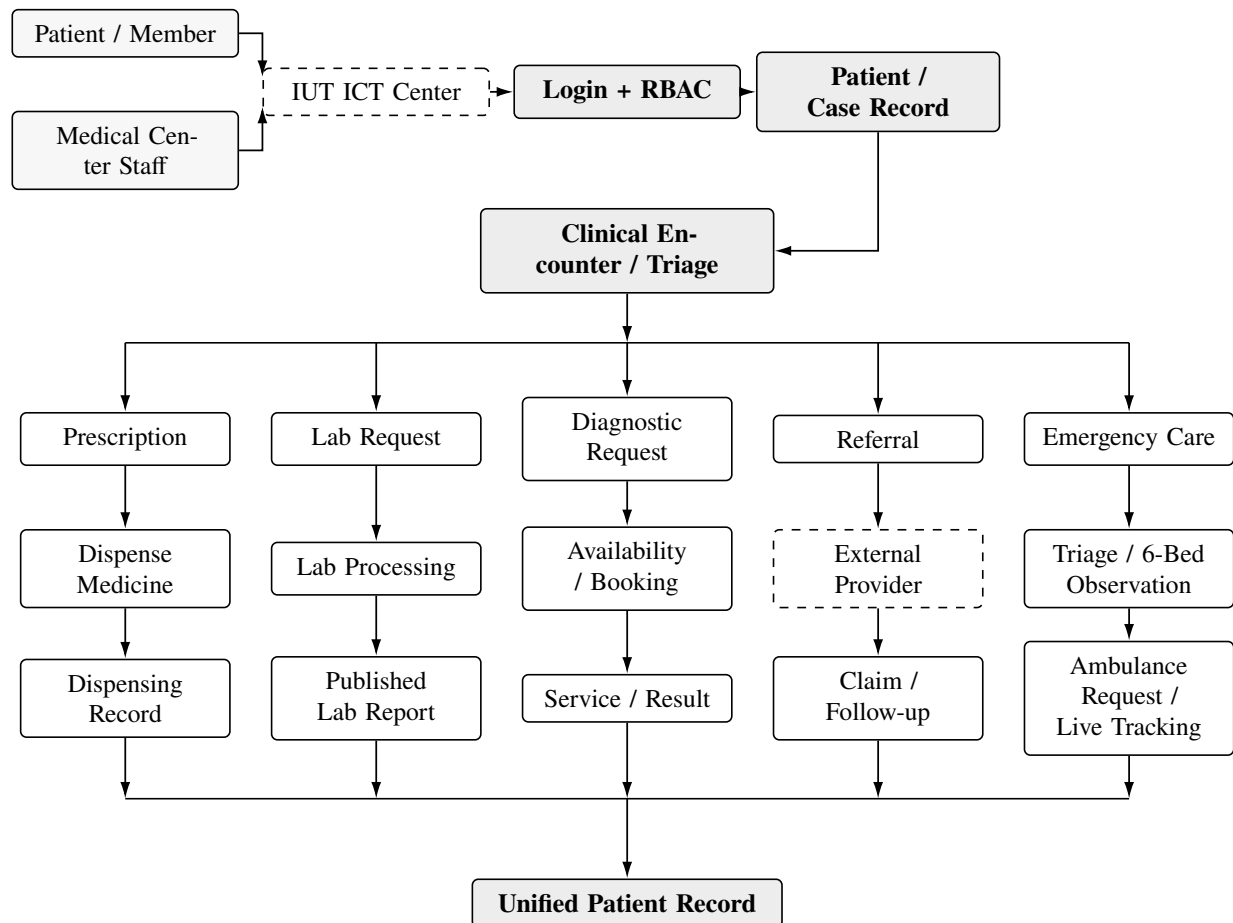


Figure 3: Core clinical and service workflow.

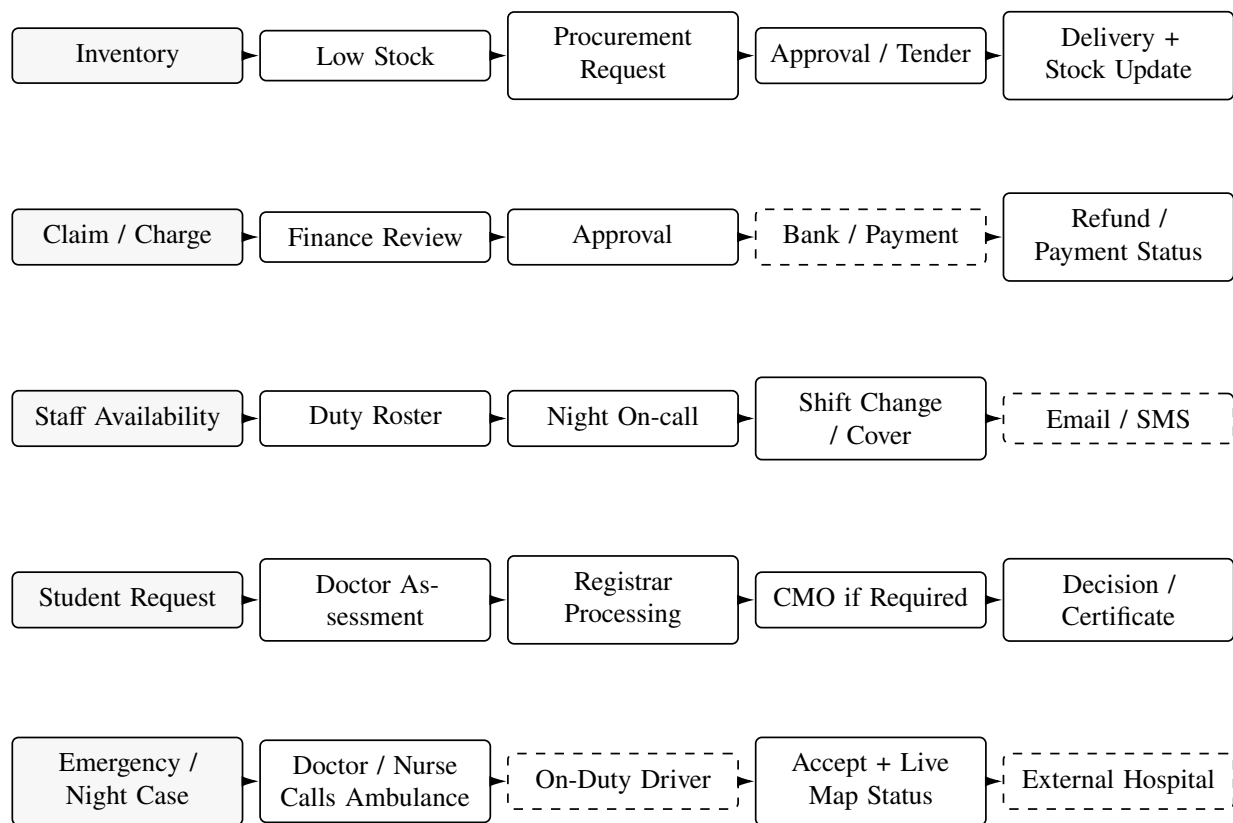


Figure 4: Supporting and external workflows.

From	To	Main Data / Event
IUT ICT Center	Medical Center Authentication / Shared Security	Institutional identity, official email, account status, user category, authorization attributes, and dependent relationships. ICT provisions accounts; the Medical Center does not.
Patient Care & Medical Record Management	Pharmacy, Inventory & Laboratory Management	Final prescription and patient identifier for dispensing; doctor-issued formal laboratory request for EPIC-17 processing.
Pharmacy, Inventory & Laboratory Management	Patient Care & Medical Record Management	Dispensing record and verified/published laboratory report document linked back to the unified patient record for authorized view/download.
Patient Care & Medical Record Management	Emergency, Diagnostic & Referral Services	Patient identity, relevant medical history, and clinical context needed for emergency, referral, or diagnostic-service workflows.
Emergency, Diagnostic & Referral Services	Financial, Billing & Reimbursement Management	Referral/diagnostic charge, payment request, receipt/refund status, and related transaction references.
Pharmacy, Inventory & Laboratory Management	Administrative, Reporting & Scheduling System	Authorized EPIC-17 blood/urine service-count data for EPIC-26 reporting; individual laboratory results are not copied into the reporting dataset.
External Hospital / Partner	Authorized Internal IUT Finance / Medical Center Staff	Referral bill/claim and supporting documents received through approved external channels; no external-hospital system login is provided.
Pharmacy, Inventory & Laboratory Management	Financial, Billing & Reimbursement Management	CMO-submitted procurement request from US-16.2, including required items and quantities.
Financial, Billing & Reimbursement Management	Pharmacy, Inventory & Laboratory Management	Approved procurement outcome and accepted delivery quantities for one-time inventory update after US-21.4.
Administrative, Reporting & Scheduling System	Emergency, Diagnostic & Referral Services	Night-call transfer handoff from US-25.2 to the existing US-7.2/US-7.3 doctor/nurse ambulance-request and driver live-tracking workflow.
External Transport Office / Ambulance Driver	Emergency, Diagnostic & Referral Services	Read-only duty availability identifies the eligible driver; the driver receives the doctor/nurse request directly and, after acceptance, provides live location/status back to authorized clinical staff. Administrative does not maintain a separate ambulance process.
Financial, Billing & Reimbursement Management	Administrative, Reporting & Scheduling System	Authorized reimbursement records read for the EPIC-27 yearly policy-adherence audit.

From	To	Main Data / Event
Administrative, Reporting & Scheduling System	Medical Staff / Patients / Shared Notification Services	Published duty/on-call schedules, personal schedule views, public on-duty information, duty-change notifications, and night-call status.
Nurse / Medical Attendant Operations	CMO / Destination Department	Submitted operational record/report reference, submitter, timestamp, CMO review decision, selected destination department, forwarding status, and routing history.
All Subdomains	Shared Services	Authentication, document storage, notification, export, audit, backup, configuration, and monitoring services used by the documented workflows.

Appendix B: TBD and Source-Gap Register

ID	Location	Description	Status
TBD-01	Technology	Final technology stack, hosting, database, and deployment approach.	Open
TBD-02	Diagnostics	Exact on-site diagnostic test list and equipment schedule ownership.	Open
TBD-03	Emergency	Bed Rest / Observation capacity confirmed as six beds for this system scope.	Resolved
TBD-04	Emergency	Verified emergency hotline number.	Open
TBD-05	Transport	Exact mobile/GPS location provider, map service, update interval, and integration protocol for the required live ambulance tracking capability.	Open
TBD-06	Integration	Final Pharmacy, Inventory & Laboratory Management inventory deduction and laboratory scheduling API contracts.	Open
TBD-07	Integration	Final Financial, Billing & Reimbursement Management payment gateway, bank API, and compliance handoff.	Open
TBD-08	Diagnostics	Ownership and maintenance process for the external diagnostic-lab directory.	Open
TBD-09	Finance / Governance	Confirm the designated higher financial authority responsible for approving disputed or over-limit refunds.	Open

ID	Location	Description	Status
GAP-01	Financial, Billing & Reimbursement Management Backlog	Financial, Billing & Reimbursement source backlog lists US-18.4, but no detailed US-18.4 story was supplied.	Source missing
GAP-02	Financial, Billing & Reimbursement Management Backlog	Financial, Billing & Reimbursement source backlog lists US-19.3, but no detailed US-19.3 story was supplied.	Source missing
GAP-03	Patient Care & Medical Record Management Criteria	Patient Care & Medical Record Management supplied detailed preconditions/postconditions and DoD but no separate acceptance-criteria section; criteria in this document are marked Derived.	Derived
GAP-04	Pharmacy/Laboratory and Finance Preconditions	Several Pharmacy, Inventory & Laboratory Management and Financial, Billing & Reimbursement Management stories omitted explicit preconditions/postconditions; conservative entries are marked Derived.	Derived